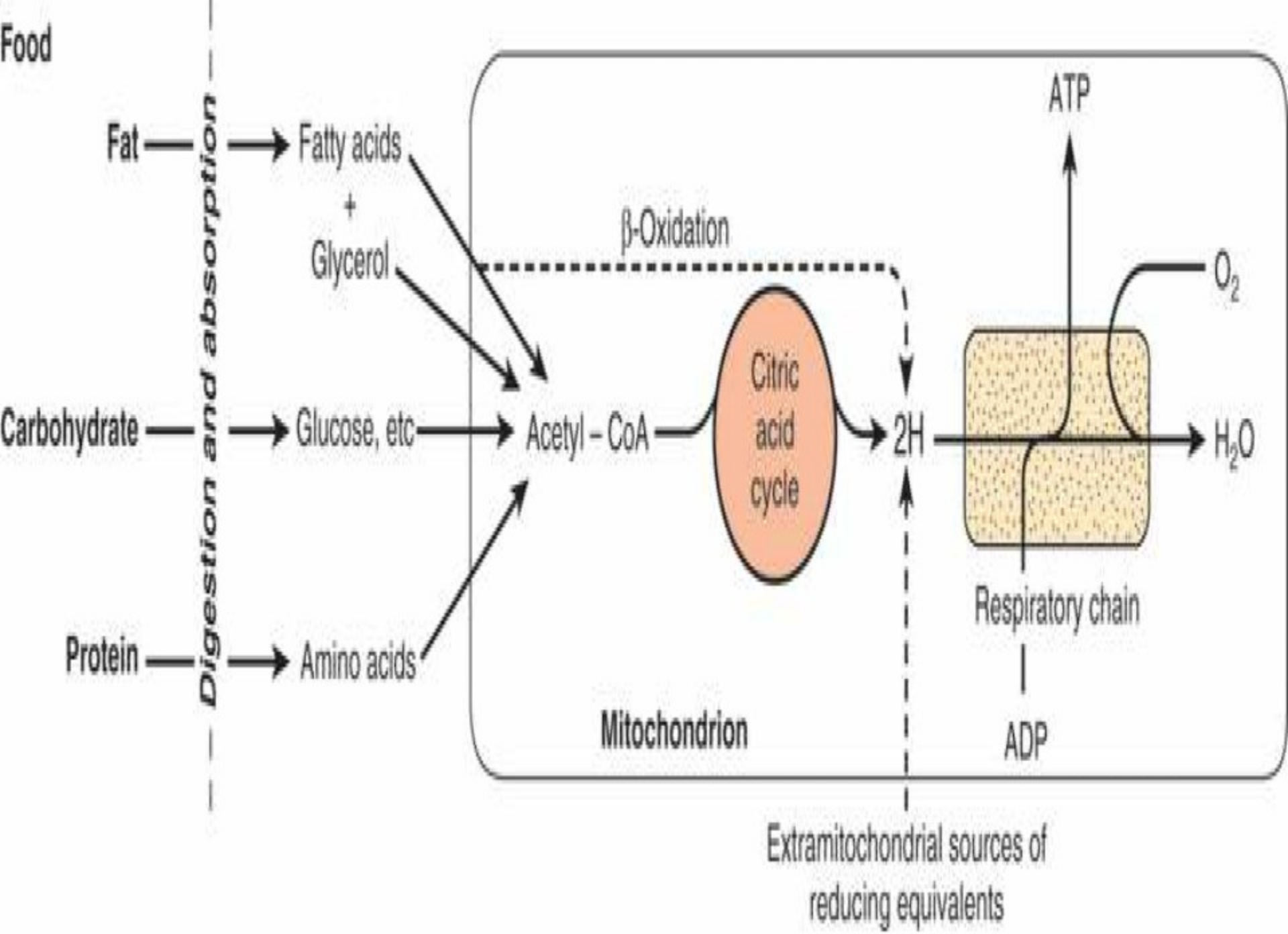
Biological Oxidation

Learning Objectives

At the end of this lecture, students should be able to learn different concepts of:

1. Biological Oxidation
2. Electron Transport Chain
3. Oxidative Phosphorylation


Food
Digestion and absorption
Fat
Fatty acids
+
Glycerol
Carbohydrate
Glucose, etc
Protein
Amino acids
β-Oxidation
Acetyl – CoA
Citric acid cycle
2H
ATP
O_2
H_2O
Respiratory chain
ADP
Mitochondrion
Extramitochondrial sources of reducing equivalents

Bioenergetics

- Bioenergetics or biochemical thermodynamics, is the study of the energy changes accompanying biochemical reactions.
- Three fundamental thermodynamic variables:
  - Enthalpy (H):
- The heat content of physical object or body (system)
- Derived from first law of thermodynamics.

- Change in enthalpy (ΔH) (Kcal/mol) is the heat absorbed or released during a reaction.
- Enthalpy is a isothermic reaction.
- Heat is not used to perform the work.
- Entropy (S):
- The randomness or disorder of a system.
- Derived from second law of thermodynamics.

- **Change in entropy (ΔS)** is the degree of randomness or disorders created during the reaction.
- **Free energy (G):**
- The maximum usable work that can be obtained from a system at constant pressure, temperature and volume.

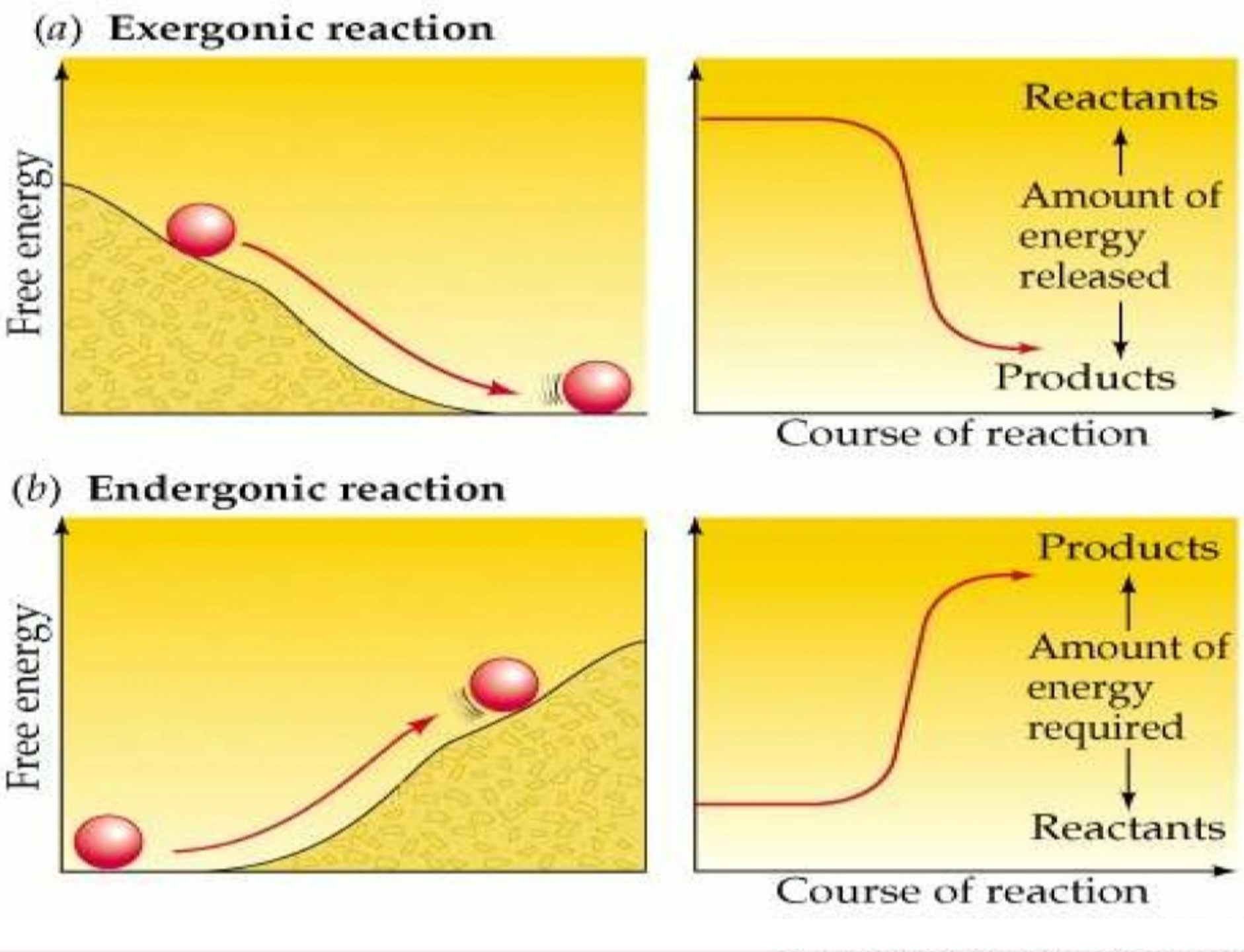
- **Free energy change (ΔG)** is the change in free energy occurring during biological reactions.
- It is related to enthalpy & entropy
- **Change in free energy can be expressed as:**
- $\Delta G = \Delta H - T \Delta S$
- ΔH is the change in enthalpy
- ΔS is the change in entropy
- T is the absolute temperature

- **Standard free energy change (ΔG):**
- **It is defined as free energy change under standard conditions.**
- **Standard condition is defined as pH 7.0, temperature 25 ℃, all reactant concentration at 1m conc, all gases at pressure 1 atmosphere.**

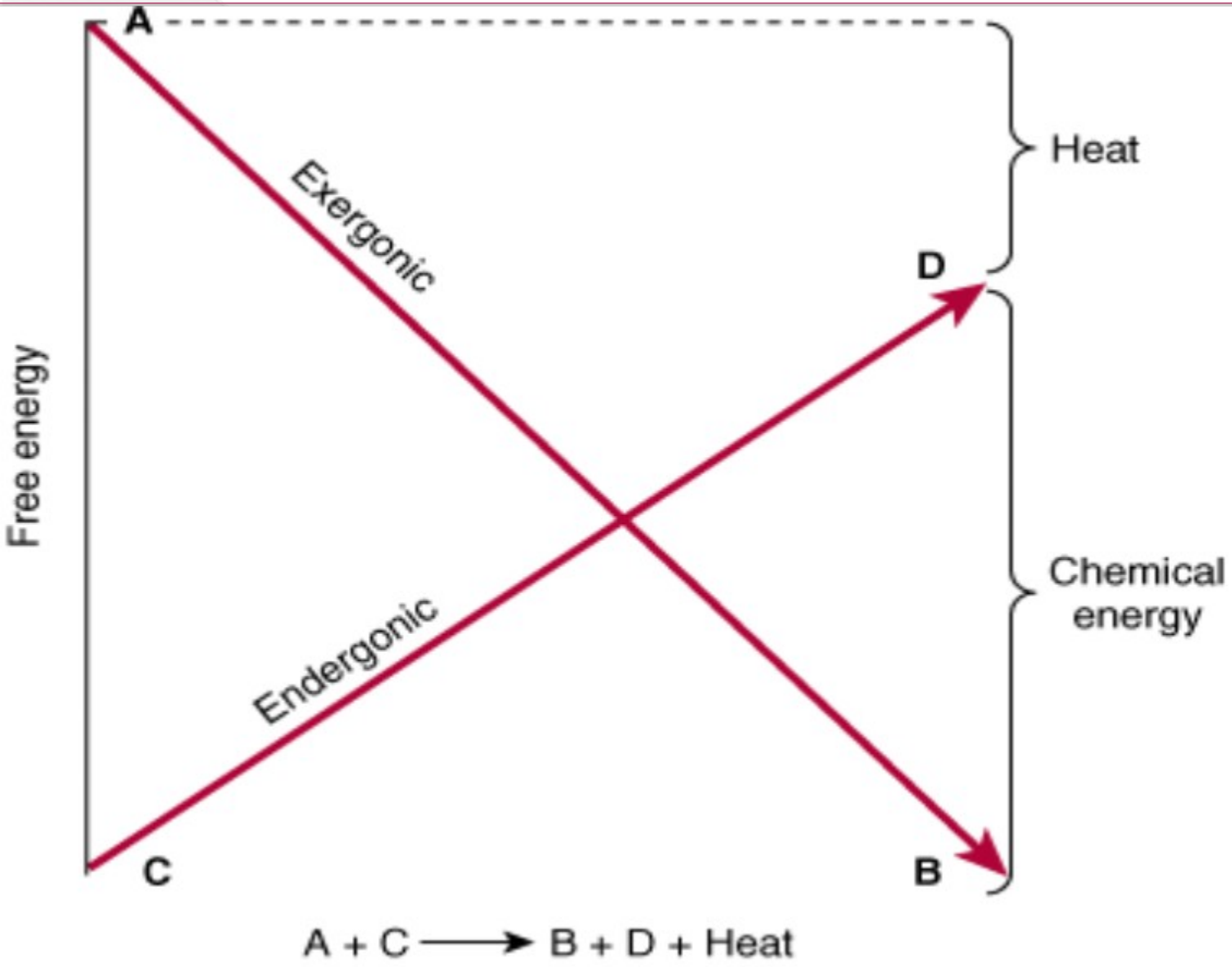
Exergonic & Endergonic Reactions

- **Exergonic reactions:**
- If the free energy change ΔG is negative in sign, the reaction proceeds spontaneously with loss of free energy & it is exergonic
- Exergonic is usually by breaking the bonds.

- **Endergonic reactions:**
- **If the free energy change ΔG is positive, the reaction proceeds only if free energy can be gained & it is endergonic.**
- **Endergonic is usually by formation of the bonds.**
- **Reactions at equilibrium: If the free energy change is zero, the reaction is at equilibrium.**


(a) Exergonic reaction
Free energy
Reactants
Amount of energy released
Products
Course of reaction
(b) Endergonic reaction
Free energy
Products
Amount of energy required
Reactants
Course of reaction

Coupling of exergonic to endergonic reaction



- If the reaction go from left to right, then the overall process must be accompanied by loss of free energy as heat.
- One possible mechanism of coupling could be envisaged if a common obligatory intermediate (I) took part in both reactions,

$$A + C \rightleftharpoons I \rightleftharpoons B + D$$

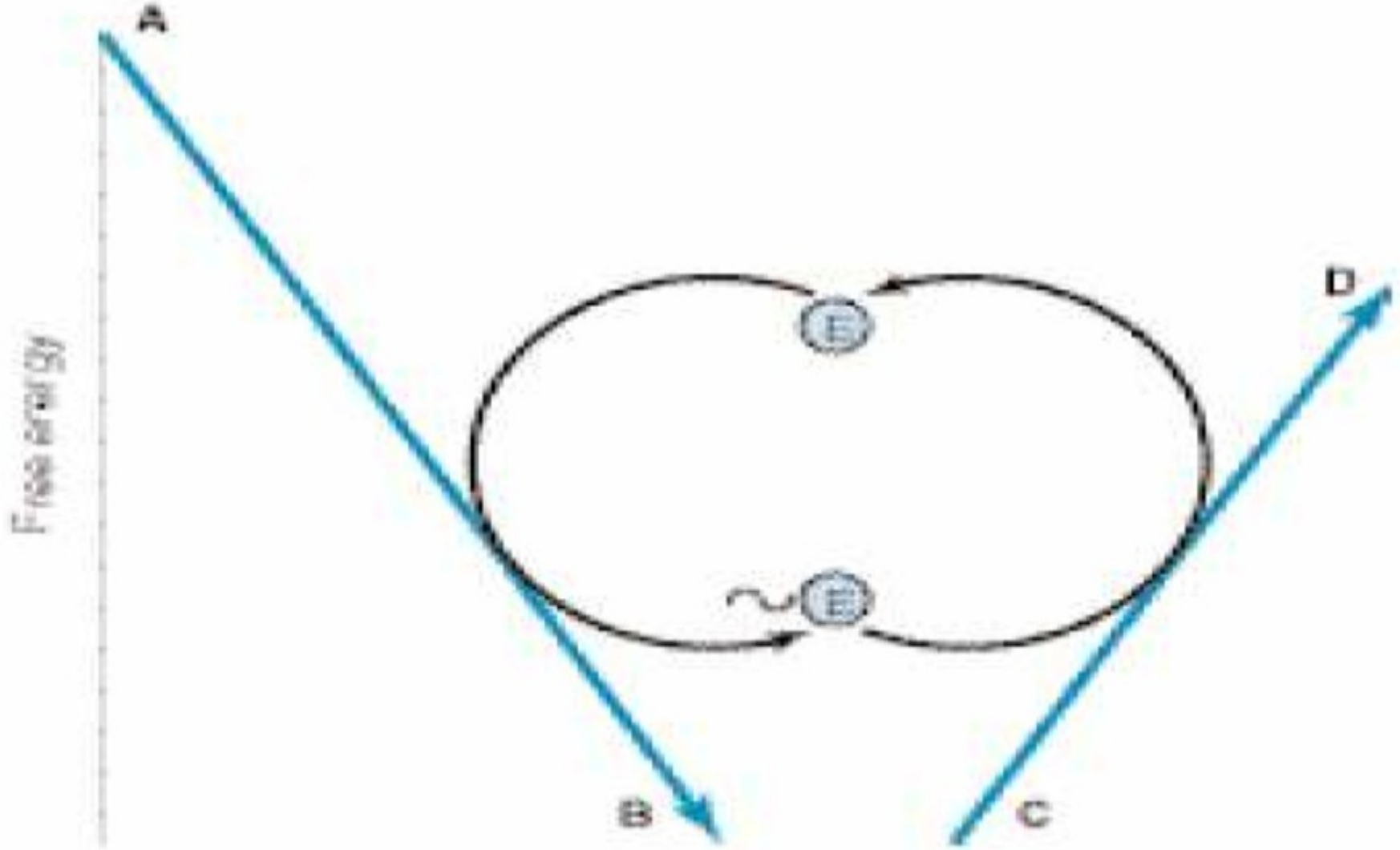


Figure 10–3. Transfer of free energy from an exergonic to an endergonic reaction via a high-energy intermediate compound (~Ⓔ).

Redox Potential

- When a substance exists both in the reduced state and the oxidized state, the pair is called a REDOX COUPLE.
- The redox potential of this couple is estimated by measuring the EMF of a sample half cell connected to a standard half-cell.

- When a substance has lower affinity for electrons than hydrogen it has a negative redox potential.
- Lower affinity for electrons = Neg. Redox potential.
- Electrons move always from more electronegative to electropositive.

Oxidation-Reduction Reactions

- Oxidation is defined as loss of electrons
- Loss of electrons occurs in three ways
  - (1) Direct loss of electrons
  - (2) Removal of hydrogen
  - (3) Addition of oxygen
- Electrons are transferred as
  - (1) Hydride ions($H^{:-}$)
  - (2) Hydrogen atoms (H)
  - (3) Electrons ($\bar{e}$)

- **Direct loss of electrons:**
- Electrons are lost directly & passed on to second acceptor molecule.
- Eg: Conversion of ferrous iron to ferric iron
- **Removal of hydrogen:**
- Electrons are lost during dehydrogenation.
- Loss of hydrogen may occur as loss of hydrogen atoms or as hydride ion which has two electrons.

Reduction:

- Reduction is defined as the gain of electrons.
- Eg: ferric iron(Fe^{3+}) to ferrous iron (Fe^{2+})

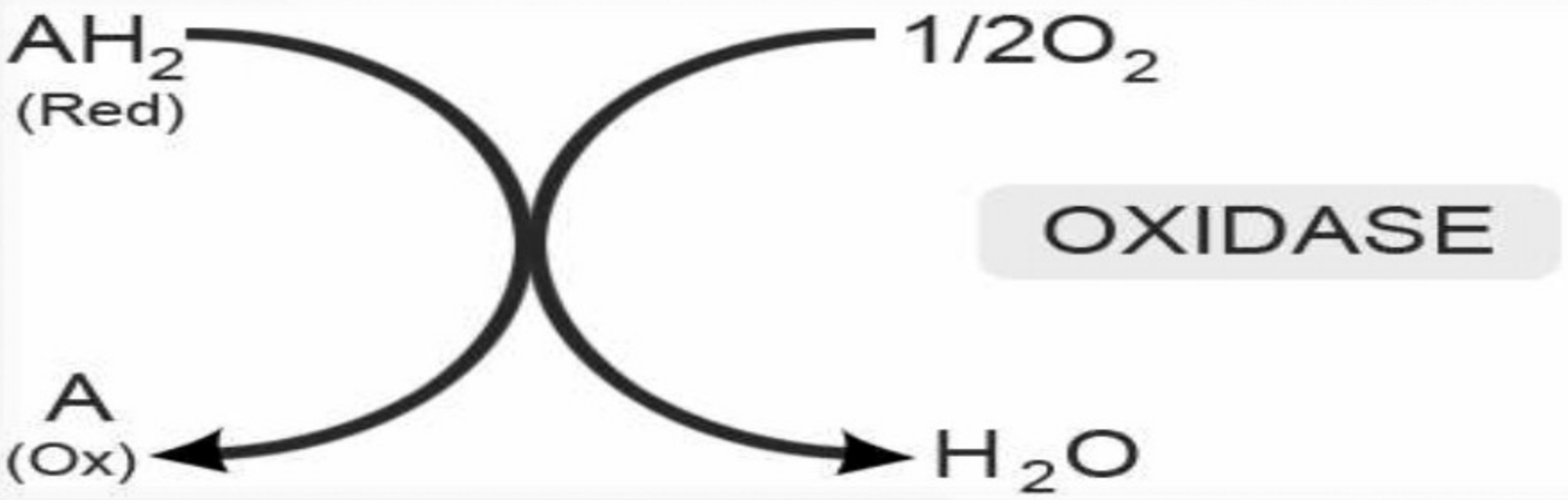
Oxidation-Reduction Reactions:

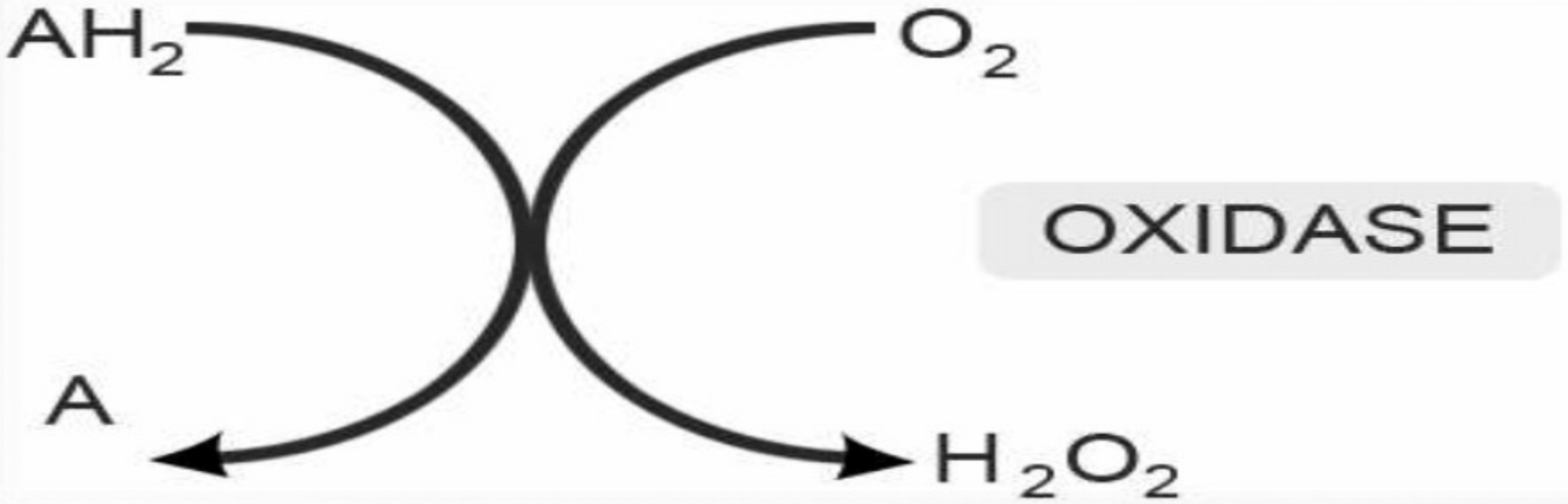
- Oxidation-reduction reactions involve transfer of electrons from one compound to another.
- When one substrate is oxidized, another substrate is simultaneously reduced.

Enzymes involved in Biological Oxidations

- **Oxidoreductases:**
- Catalyzes oxidation & reduction reactions.
- They catalyze the addition of oxygen, transfer of hydrogen & transfer of electrons.
- **Subclass:** Subclasses are oxidases, dehydrogenases, oxygenases & hydroperoxidases.
- **Oxidases:**
- Catalyze the transfer of hydrogen or electrons from the donor, **using oxygen as hydrogen acceptor.**
- The reaction product may be H_2O or H_2O_2

Oxidases



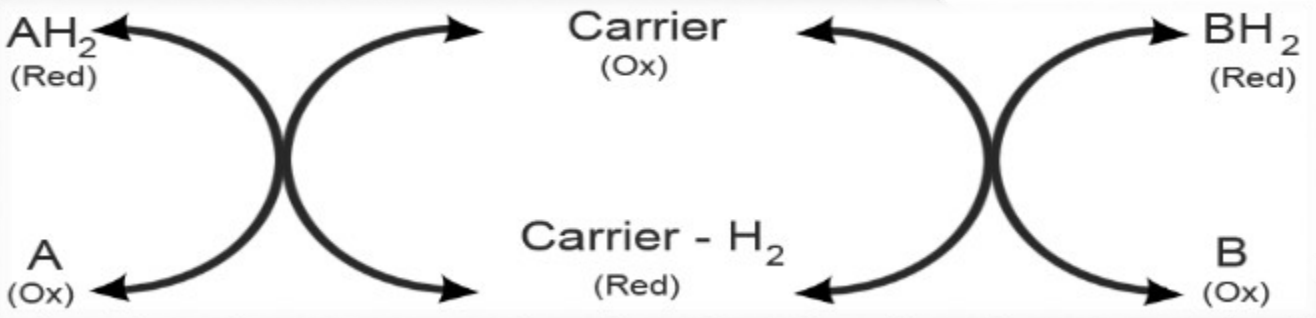


Aerobic Dehydrogenases

- They contain flavoprotein (FAD or FMN) as coenzymes.
- They transfer hydrogen atoms from the substrate to oxygen via flavin carriers.
- They are called as aerobic dehydrogenases.
- They are also capable of transferring hydrogen to acceptors other than oxygen.

Anaerobic Dehydrogenases

- Perform 2 main functions:
- Transfer hydrogen from one substrate to another in a coupled oxidation-reduction reactions.
- As components of ETC Dehydrogenases use coenzymes – nicotinamides & riboflavin - as hydrogen carriers



Dehydrogenae Specific for A

Dehydrogenae Specific for B

Types of Dehydrogenases

- Dehydrogenases;
- Catalyze the transfer of hydrogen (or electrons), but the hydrogen acceptor is molecule other than oxygen.
- The hydrogen acceptors are coenzymes,
- E.g. NAD, NADP, FAD & FMN.
- NAD linked dehydrogenases :

$$H_2 \rightarrow H + H^+ + e^-$$

$$AH_2 + NAD^+ \rightarrow A + NADH + H^+$$

- **$NADP^+$ linked dehydrogenases:**
- Reductive biosynthesis of various substances e.g Fatty acid biosynthesis
- **FAD linked Dehydrogenases:**
- FAD is the coenzymes instead of NAD. e.g Succinate dehydrogenase
- **Cytochomes:**
- All cytochromes (Except Cytochrome Oxidase) are anaerobic dehydrogenases

Oxygenases

- Oxygenases catalyze the direct incorporation of oxygen into the substrate.
- Oxygen is bound to active site of the enzyme.
- There are two types of oxygenases:
- Monooxygenases & Dioxygenases.
- Monooxygenases:
- These will catalyze the incorporation of only one oxygen to the substrate.

They are also called as hydroxylases or mixed function oxygenases.

Phenylalanine + O_2 + Biopterin → Tyrosine + H_2O + Dihydrobiopterin

Dioxygenases: These will catalyze the incorporation of both atoms of oxygen into the substrate.

Homogenstisic acid + O_2 → Maleylacetoacetate

- **Hydroperoxidases:**
- **These enzymes will utilize hydrogen peroxide as (H_2O_2) as the substrate.**
- **These are two types:**
- **Peroxidases:**
- **Catalase:**

Peroxidases:

- Utilize H_2O_2 as oxygen donor but O2 acceptor is a molecule other than H_2O_2.
- Eg. Glutathione peroxidase.

Catalase:

- It is a unique enzyme & utilizes H_2O_2 as both donor & acceptor of oxygen (electrons).

E.g: $H2O2 + H2O2 \longrightarrow 2H2O + O2$

- Catalase functions in the cell to detoxify H_2O_2.
- Peroxisomes are rich in oxidases and catalases.
- Coenzymes involved in Biological Oxidations are:
- NAD^+, $NADP^+$, FAD^+, FMN^+

High energy compounds

- Certain compounds are encountered in the biological system which , yield energy.
- Energy rich compounds or high-energy rich compounds is substances which possess sufficient free energy to liberate at least 7 Cal/mol at pH 7.0

- Certain other compounds which liberate less than 7.0 cal/mol.
- Are referred to as low energy compounds
- Indicated by SQUIGGLE bond (~)
- Free energy varies from -7 to -15 kcal/mol

Compound	$\Delta G^{0'}$	
	kJ/mol	kcal/mol
Phosphoenolpyruvate	–61.9	–14.8
Carbamoyl phosphate	–51.4	–12.3
1,3-Bisphosphoglycerate (to 3-phosphoglycerate)	–49.3	–11.8
Creatine phosphate	–43.1	–10.3
$ATP \rightarrow ADP + P_i$	–30.5	–7.3
$ADP \rightarrow AMP + P_i$	–27.6	–6.6
Pyrophosphate	–27.6	–6.6
Glucose 1-phosphate	–20.9	–5.0
Fructose 6-phosphate	–15.9	–3.8
AMP	–14.2	–3.4
Glucose 6-phosphate	–13.8	–3.3
Glycerol 3-phosphate	–9.2	–2.2

Classification of high energy compounds

- There are at least 5 groups of high energy compounds
- Pyrophosphates, eg, ATP
- Acyl phosphates, eg,1,3-bisphosphoglycerate
- Enol phosphates, eg, PEP
- Thioesters, eg, Acetyl CoA
- Phosphagens, eg, Phosphocreatine

High-energy bonds

- The high-energy compounds possess acid anhydride bonds (mostly phosphoanhydride bonds) which are formed by the condensation of two acidic groups or related compounds.
- These bonds are referred as high-energy bonds.
- Free energy is liberated when these bonds are hydrolysed.
- ATP is most important high-energy compound

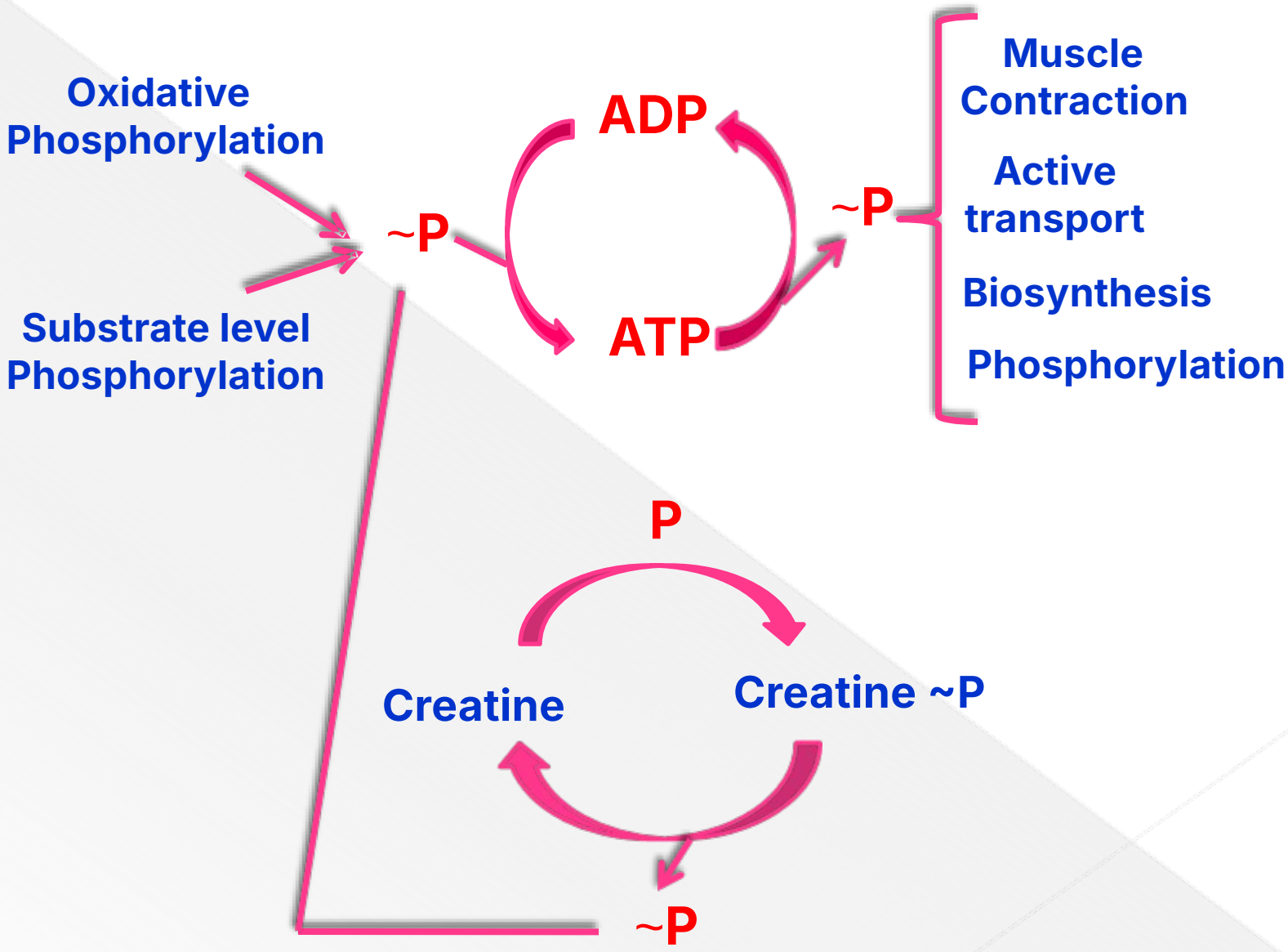
ATP-ADP Cycle

- The hydrolysis of ATP is associated with the release of large amount of energy.

$$ATP + H_2O \longrightarrow ADP + Pi + 7.3\ Cal$$

- The energy liberated is utilized for various process like muscle contraction, active transport etc.
- ATP can also acts as a donor of high-energy phosphate to low-energy compounds, to make them energy rich.
- ADP can accepts phosphate to form ATP.

ATP-ADP cycle along with sources & utilization of ATP



- ATP serves as an immediately available energy currency of the cell which is constantly being utilized & regenerated.
- ATP acts as an energy link between the catabolism & anabolism in the biological systems.
- Hydrolysis of ATP releases 7.3kcal/mol.

- At rest, Na ± K ± ATPase uses up one-third of all ATP formed.
- An average person at rest consumes & regenerates ATP at a rate of approximately 3 molecules per second, i.e. about 1.5 kg/day.

Synthesis of ATP

- ATP can be synthesized in two ways
- **Oxidative phosphorylation:**
- Major source of ATP in aerobic organisms.
- It is linked with mitochondrial ETC.
- **Substrate level phosphorylation:**
- When the energy of high energy compound is directly transferred to nucleoside diphosphate to form a triphosphate without the help from ETC.

- The high-energy compounds such as
  - PEP
- 1,3-bisphosphoglycerate
- Succinyl CoA can transfer high-energy phosphate to ultimately produce ATP.

- **Storage forms:**
- **Phosphocreatine (creatine phosphate) provides high energy reservoir of ATP to regenerate ATP rapidly, catalyzed by creatine kinase.**
- **Stored mainly in muscle & brain.**
- **In invertebrates, phosphoarginine (arginine phosphate) is storage form.**

Biological oxidation

- The transfer of electrons from the reduced coenzymes through the respiratory chain to oxygen is known as biological oxidation.
- Energy released during this process is trapped as ATP.
- This coupling of oxidation with phosphorylation is called oxidative phosphorylation.

Redox potentials

- **Oxidation:**
- Oxidation is defined as the loss of electrons and reduction as the gain in electrons.
- When a substance exists both in the reduced state & in the oxidized state, the pair is called a redox couple.

Redox potential(E_0):

- The oxidation-reduction potential or redox potential, is a quantitative measure of the tendency of a redox pair to lose or gain electrons.
- The redox pairs are assigned specific standard redox potential at pH 7.0 & 25^0 C

Standard redox potential (E_0) of some oxidation-reduction systems

Redox pair	E_0 Volts
Succinate/α -ketoglutarate	-0.67
$2H^+/H_2$	-0.42
NAD^+/NADH	-0.32
FMN/$FMNH_2$	-0.30
Lipoate (ox/red)	-0.29
FAD/$FADH_2$	-0.22
Puruvate/lactate	-0.19
Fumarate/succinate	+0.03
Cytochrome b (Fe^{3+}/Fe^{2+})	+0.07
CoenzymeQ (ox/red)	+0.10
Cytochrome c_1 (Fe^{3+}/Fe^{2+})	+0.23
Cytochrome c (Fe^{3+}/Fe^{2+})	+0.25
Cytochrome a (Fe^{3+}/Fe^{2+})	+0.29
½ O_2/H_2O	+0.82

- The more negative redox potential represents a greater tendency to lose electrons.
- A more positive redox potential indicates a greater tendency to accept electrons
- The electrons flow from a redox pair with more negative E_0 to another redox pair with more positive E_0
- The redox potential (E_0) is directly related to the change in the free energy (ΔG^0)

Transport of Reducing Equivalents- Shuttle Pathways

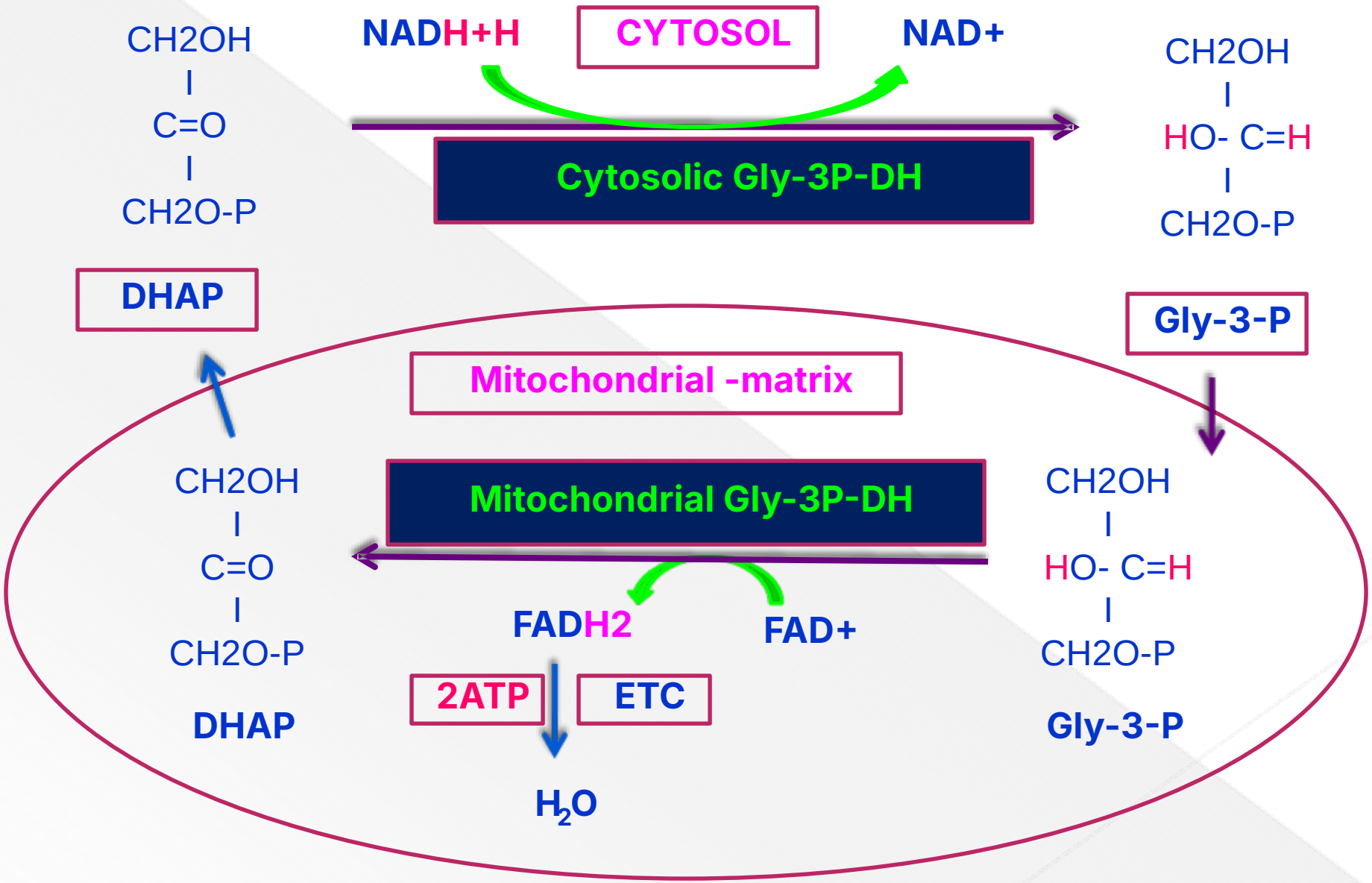
- The inner mitochondrial is impermeable to NADH.
- Therefore, the NADH produced in the cytosol cannot directly enter the mitochondria.
- Two pathways
- Glycerol-phosphate shuttle
- Malate-aspartate shuttle

Glycerol-phosphate shuttle

- Cytosolic glycerol 3-phosphate dehydrogenase oxidizes NADH to NAD^+
- The reducing equivalents are transported through glycerol 3-phosphate into the mitochondria.
- Glycerol 3-phosphate dehydrogenase-present on outer surface of inner mitochondrial membrane – reduces FAD to $FADH_2$.

- Dihydroxyacetone phosphate (DHAP) escapes into the cytosol & the shuttling continues.
- $FADH_2$ gets oxidized via ETC to generate 2ATP

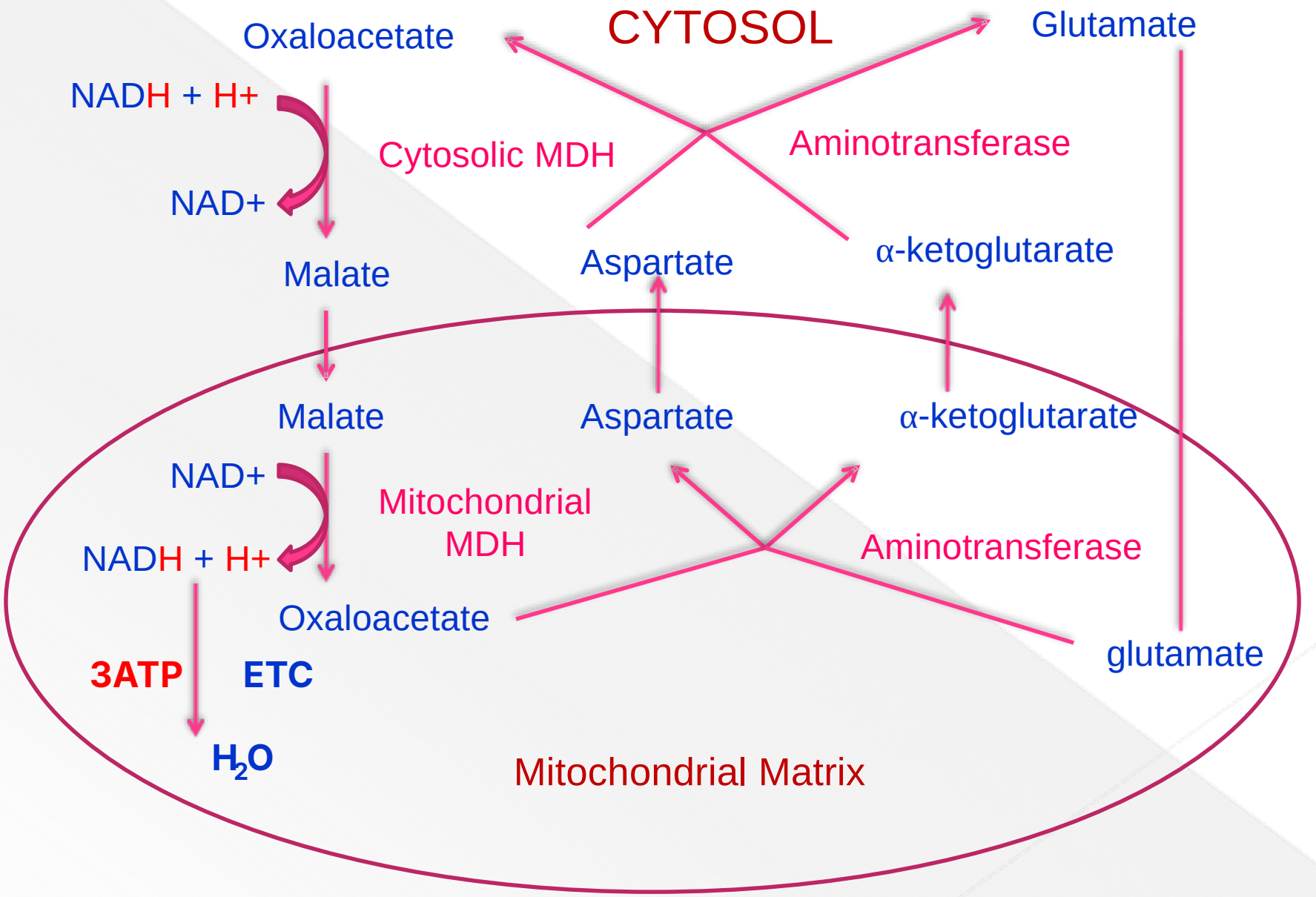
Glycerol-phosphate -shuttle



Malate-aspartate shuttle

- In the cytosol, oxaloacetate accepts the reducing equivalents (NADH) & becomes malate.
- Malate enters the mitochondria where it is oxidized by mitochondrial MDH
- In this reaction, NADH & oxaloacetate are regenerated.
- NADH gets oxidized via ETC & 3 ATP are produced.

Malate-aspartate shuttle



- In the mitochondria, oxaloacetate participates in transamination reaction with glutamate to produce aspartate & α-ketoglutarate.
- The aspartate enters the cytosol & transaminates with α-ketoglutarate to give oxaloacetate & glutamate.

Electron transport chain or respiratory chain

- The flow of electrons occurs through successive dehydrogenase enzymes in mitochondria , together known as the ETC.

  (the electrons are transferred from higher to lower potential.)

  - **Significance:**
  - The free energy released during the transport of electrons is utilized for the formation of ATP.

Mitochondrial organization

- Mitochondria consists of five distinct parts
  - Outer membrane, inner membrane, intermembrane space, cristae & matrix
  - **Inner mitochondrial membrane:**
- The ETC & ATP synthesizing system are located on inner mitochondrial membrane, which is specialized structure, rich in proteins.

- Inner membrane is highly folded to form cristae.
- Surface area of inner mitochondrial membrane is increased due to cristae.
- The inner surface of inner mitochondrial membrane possesses specialized particles, the phosphorylating subunits which are centres for ATP production.

Components

- ETC consists of four enzymes complexes & two free electron carriers.
  - Enzyme complexes:
  - ComplexI: NADH-ubiquinone oxido-reductase
  - Complex II: Succinate dehydrogenase
  - Complex III: Ubiquinol cytochrome oxido-reductase

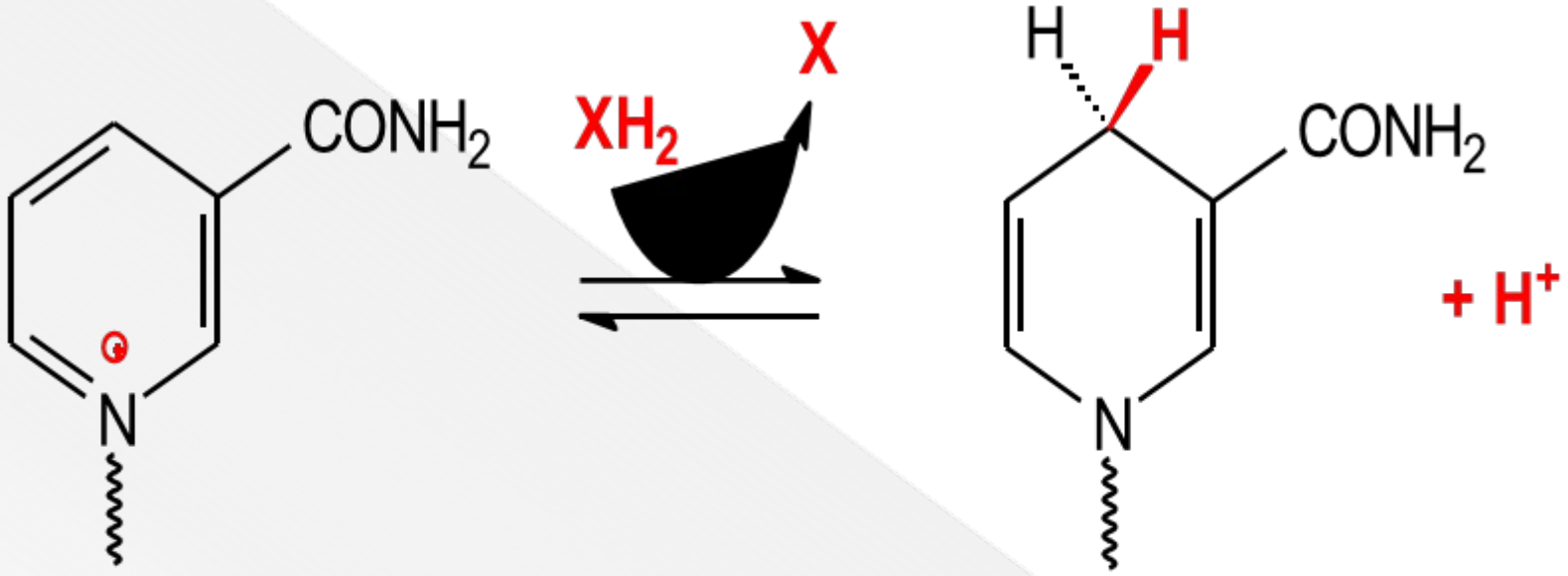
- **Complex IV:** Cytochrome oxidase
- Two free electron carriers are coenzyme Q & Cytochrome C.
- **Complex V:** It is ATP synthase.
- The complexes I-IV are carriers of electrons while complex V is responsible for ATP synthesis.

- The enzyme complexes & mobile carriers are collectively involved in the transport of electrons which, ultimately, combine with oxygen to produce water.
- Largest proportion of O_2 supplied to body is utilized by mitochondria for the operation of ETC.

Complex I

- Of the two coenzymes NAD^+ & $NADP^+$, NAD^+ is more actively involved in ETC.
- Tightly bound to the inner membrane
- NAD is reduced to NADH+ H^+ by dehydrogenases with the removal of two hydrogen atoms from the substrates, the substrates includes pyruvate, gly-3-P. etc.
- NADPH is more effectively utilized for anabolic reactions - fatty acid synthesis, cholesterol synthesis.

Nicotinamide coenzymes: NAD^+

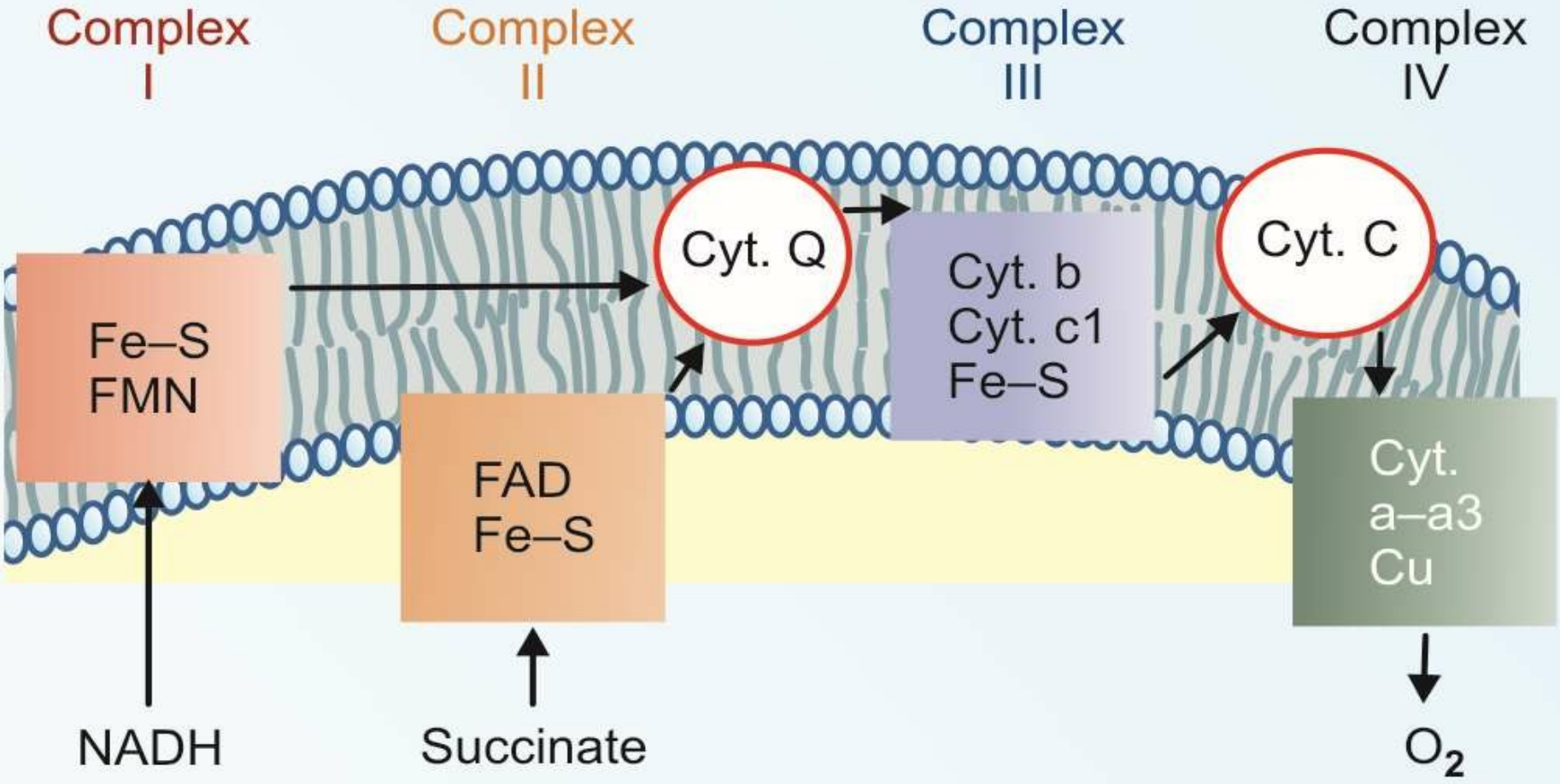


oxidised coenzyme

NAD^+ or $NADP^+$

reduced coenzyme

NADH or NADPH



Complex I = NADH-CoQ reductase (NADH dehydro-genase complex). Complex II = Succinate-Q-reductase. Complex III = Cytochrome reductase (Cytochrome b-c1 complex). Complex IV = Cytochrome oxidase.

Flavoproteins

- The enzyme NADH dehydrogenase (NADH-coenzyme Q reductase) is a flavoprotein with FMN as the prosthetic group.
- The coenzyme FMN accepts two electrons & a proton to form $FMNH_2$.

- **NADH dehydrogenase** is a complex enzyme closely associated with non-heme iron proteins or iron-sulfur proteins.
- In this, 4 protons are pumped out from mitochondria.

$$NADH + H^{+} + FMN \longrightarrow NAD^{+} + FMNH_2$$

COMPLEX II: Succinate-CoQ-Reductase

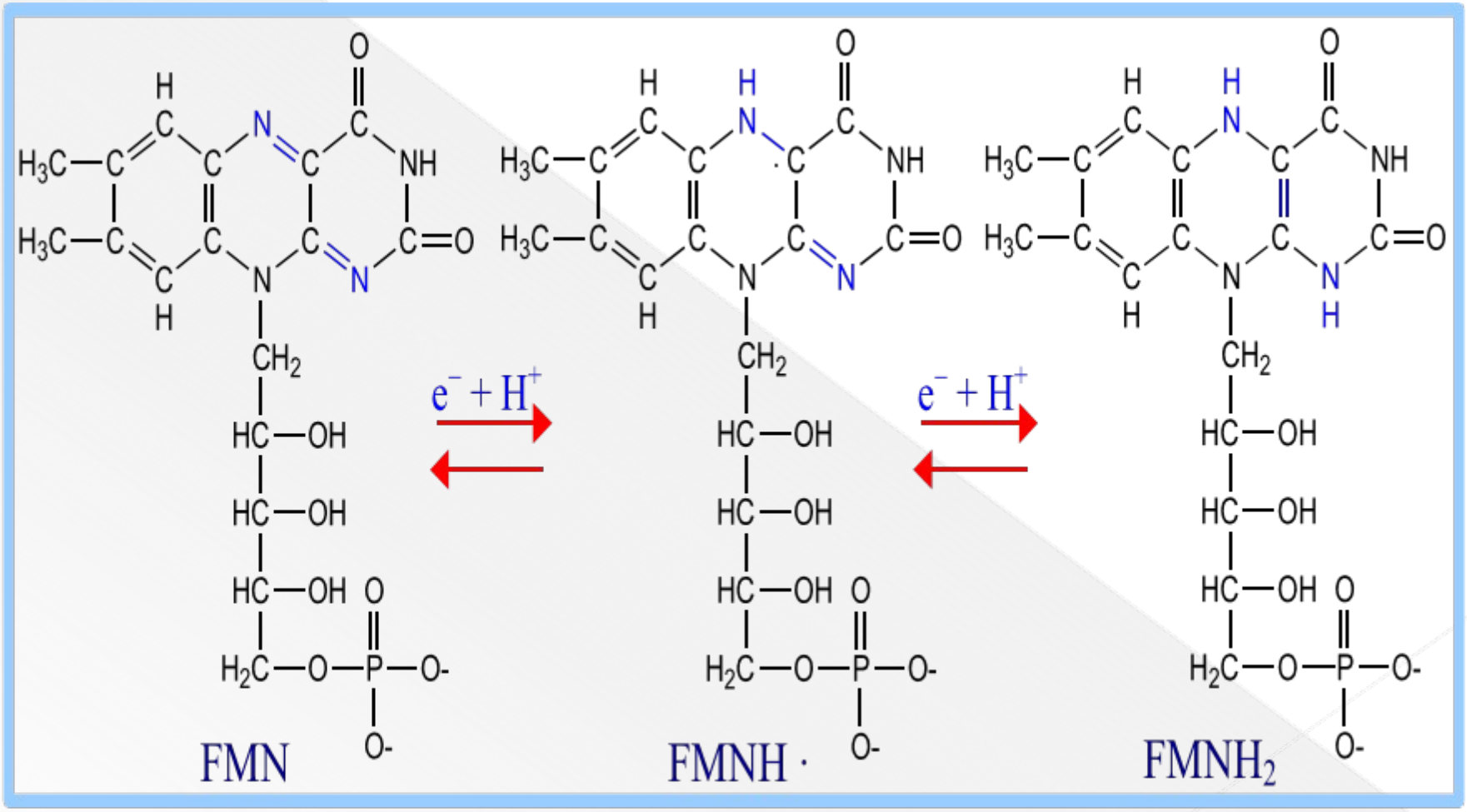
- The electrons from $FADH_2$ enter ETC at the level of Co Q.
- Succinate DH is an enzyme found in inner mitochondrial membrane.
- It is also a flavoprotein with FAD as coenzyme.
- The 3 major enzyme systems that transfer their electrons directly to ubiquinone are:

a. Succinate dehydrogenase

b. Fatty acyl CoA dehydrogenase

c. Mitochondrial glycerol phosphate dehydrogenase.

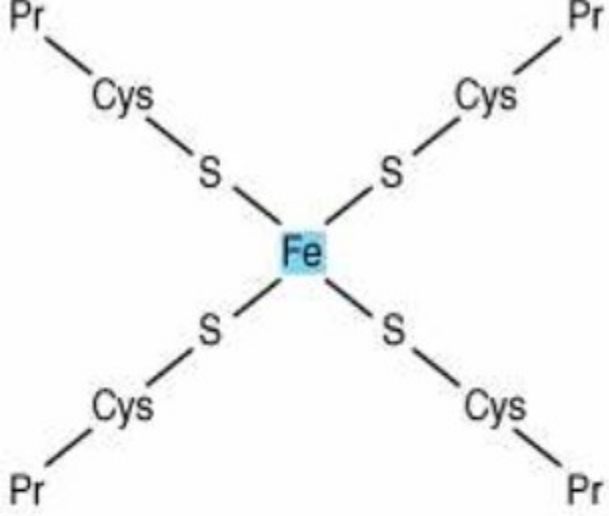
Oxidation and reduction of flavin coenzymes



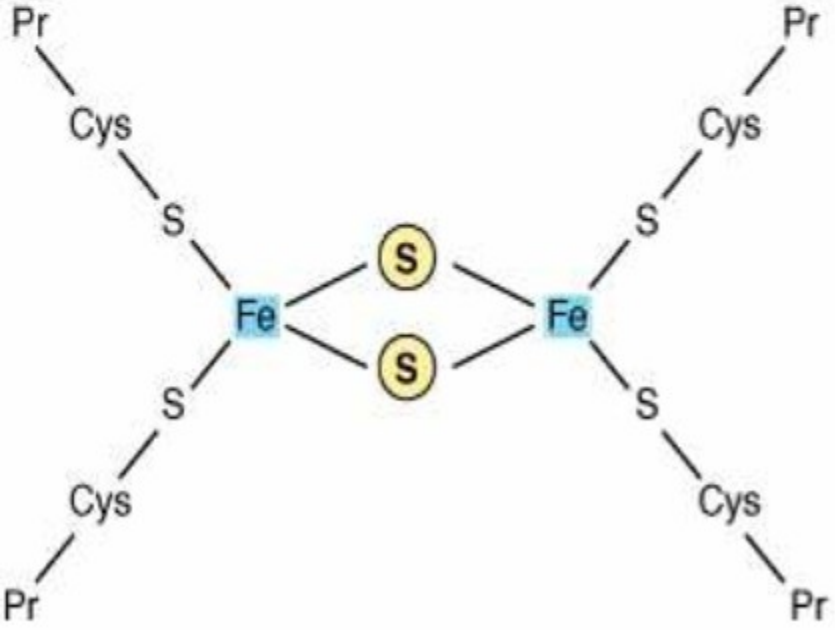
Iron-sulfur Centers (clusters)

- Iron-sulfur centers (Fe-S) are prosthetic groups containing 1-4 iron atoms
- Iron-sulfur (Fe-S) proteins exist in the oxidized (Fe^{3+}) or reduced (Fe^{2+}) state.
- Iron-sulfur centers transfer only one electron, even if they contain two or more iron atoms

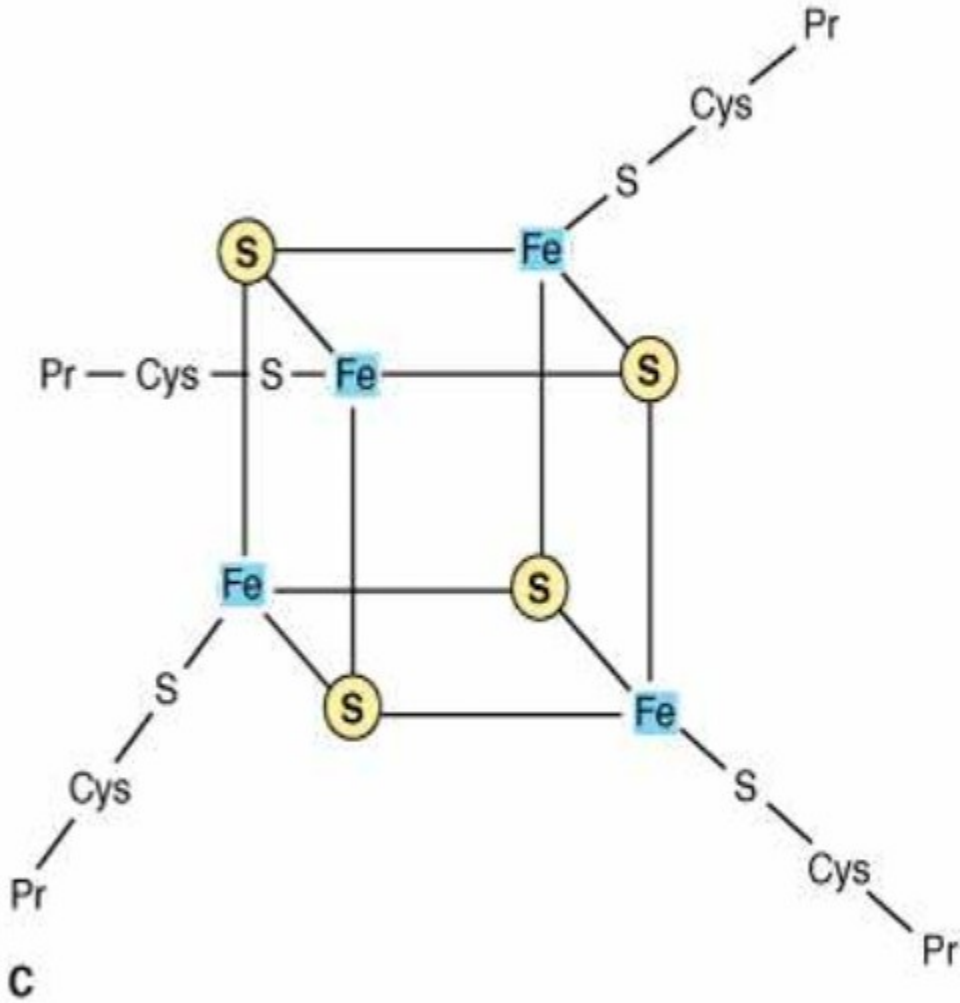
- Fe-S participates in the transfer of electrons from FMN to coenzyme Q.
- Other Fe-S proteins associated with cytochrome b & cytochrome c1 participate in the transport of electrons.


Pr
Cys
S
Fe
Pr
Cys
S
S
S
Cys
Cys
Pr
Pr

A


Pr
Cys
S
Fe
S
S
Fe
Pr
Cys
S
S
Cys
Cys
Pr
Pr

B


Pr
Cys
S
S
Fe
Pr — Cys — S — Fe
S
Fe
S
S
S
Fe
S
Cys
Pr
Cys
Pr

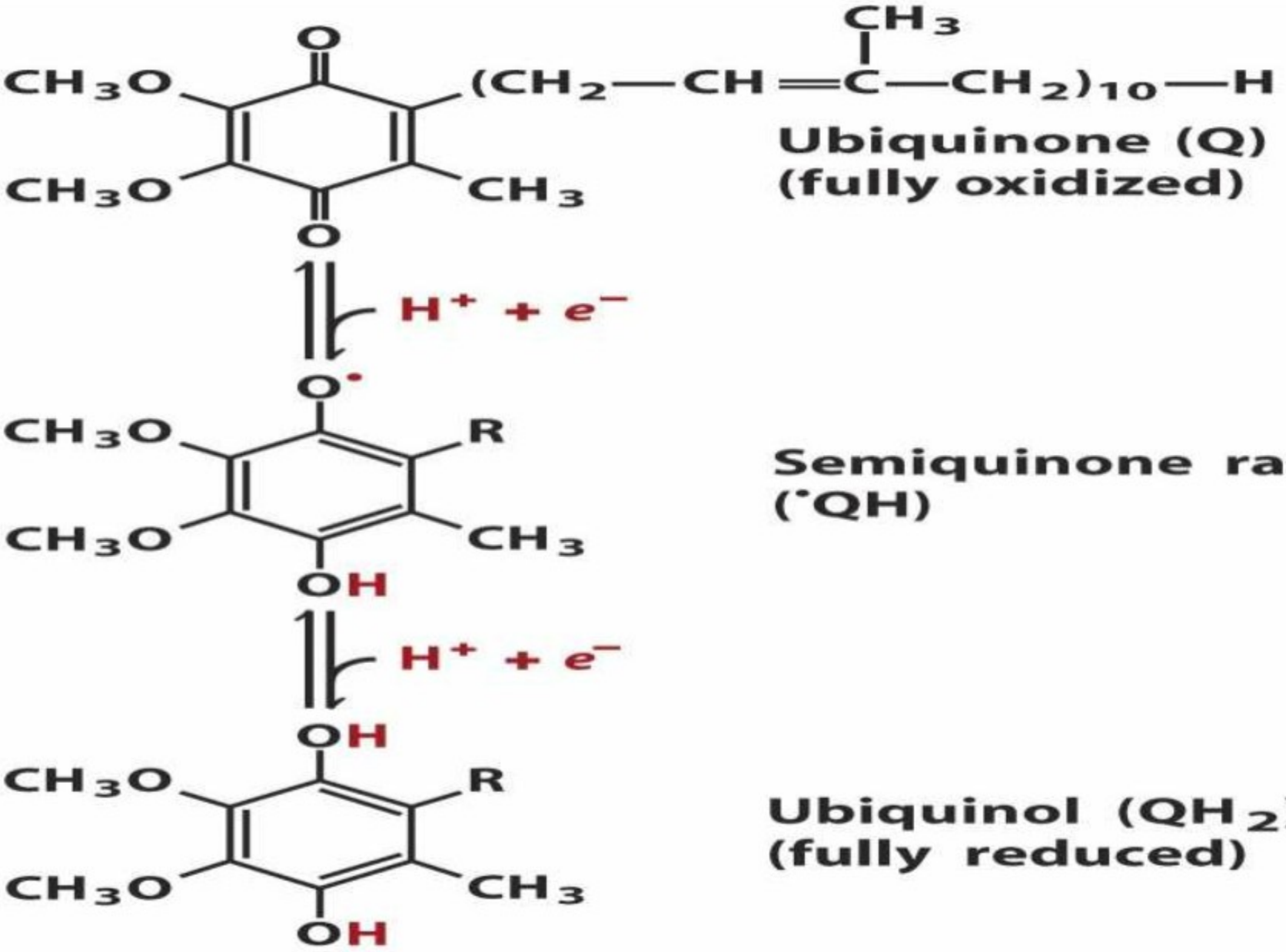
C

Coenzyme Q

- It is also known as ubiquinone.
- It is a quinone derivative with isoprenoid side chain
- Mammalian tissues possess a quinone with 10 isoprenoid units which is known as coenzyme Q_{10}
- The ubiquinone is reduced successively to semiquinone (QH) & finally to quinol (QH2)

- It accepts a pair of electrons from NADH or FADH2 through complex I or complex II respectively.
- 2 molecules of cytochrome c are reduced.
- The Q cycle facilitates the switching from the 2 electron carrier ubiquinol to the single electron carrier cytochrome c.
- This is a mobile carrier.

Coenzyme Q


CH3O
CH3O
O
O
(CH2—CH=C—CH2)10—H
CH3
CH3
Ubiquinone (Q)
(fully oxidized)
H+ + e−
O·
CH3O
CH3O
R
CH3
OH
Semiquinone radical
(·QH)
H+ + e−
OH
CH3O
CH3O
R
CH3
OH
Ubiquinol (QH2)
(fully reduced)

COMPLEX III: Cytochrome Reductase

- This is a cluster of iron-sulphur proteins, cytochrome b & cytochrome c1, both contain heme prosthetic group.
- Cytochromes are conjugated proteins
- Consists of a porphyrin ring with iron atom.
- Heme group of cytochromes differ from that found in Hb & myoglobin.

- The iron of heme in cytochromes is alternately oxidized (Fe^{3+}) & reduced (Fe^{2+})
- Which is essential for transport of electrons in the ETC.
- In this, 4 protons are pumped out.
- The electrons transported from coenzyme Q to cytochromes b, c1, c, a & a3.

- The property of reversible oxidation-reduction of heme iron present in cytochromes allows them to function as effective carriers of electrons in ETC.
- Cytochrome C:
- It is a small protein containing 104 amino acids & a heme group.
- It is a loosely bound to inner mitochondrial membrane & can be easily extracted.

COMPLEX IV: Cytochrome Oxidase (Cytochrome a & a_3)

- Contains cytochrome a and cytochrome a_3
- Which is the terminal component of ETC
- Tightly bound to inner mitochondrial membrane.
- Cytochrome oxidase is the only electron carrier, heme iron of which can directly react with molecular oxygen.

- It also contains copper that undergoes oxidation-reduction during transport of electrons.
- 2 protons are pumped out.
- In the final stage of ETC, the transported electrons, the free protons & the molecular oxygen combine to produce water.

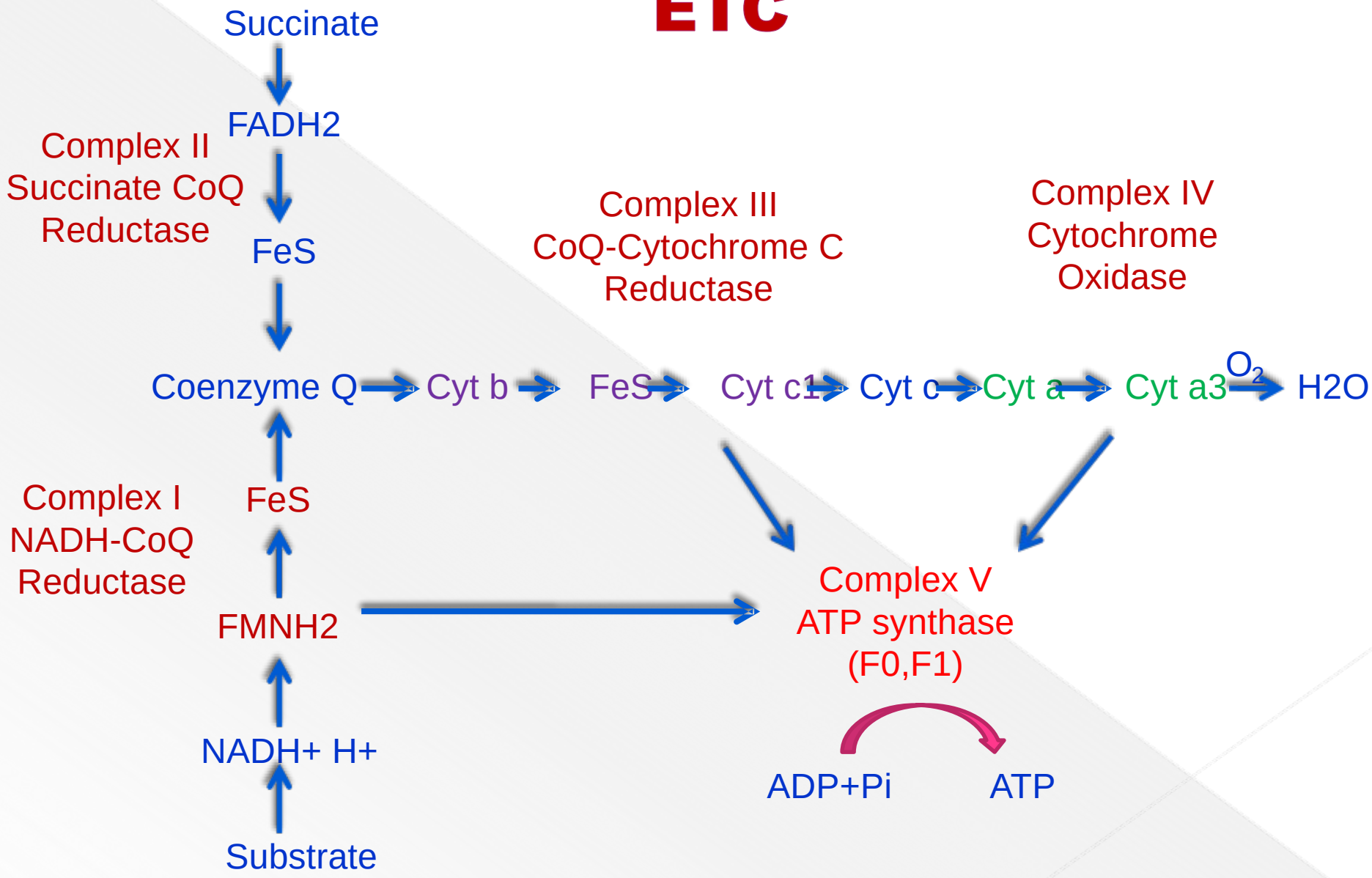
Sources of Electrons

- **Electrons donors:**
  - **NADH & $FADH_2$**
- **NADH: It is produced in the following reactions**
- **PDH complex: It transfers electrons from pyruvate to NAD^+**
- **α-ketoglutarate DH: It transfers electrons from alpha-ketoglutarate to NAD^+**

- **Isocitrate DH:** It transfers electrons from isocitrate to NAD^+
- **Malate DH:** It transfers electrons from malate to NAD^+
- **Hydroxyacyl CoA DH:** It transfers electrons from hydroxy acyl CoA to NAD^+
- **$FADH_2$**
- FAD is tightly bound to enzymes called flavoproteins.

- $FADH_2$ is produced in the following reactions.
- **Succinate DH (complex II):**
- It transfers electrons from succinate to FAD.
- **Glycerol 3-P DH:** It transfers electrons from glycerol 3-P to FAD.
- **Fatty acyl CoA DH:** It transfers electrons from fatty acids to FAD.
- $FADH_2$ donates electrons to coenzyme Q

ETC


Succinate
FADH2
Complex II
Succinate CoQ
Reductase
FeS
Complex III
CoQ-Cytochrome C
Reductase
Complex IV
Cytochrome
Oxidase
Coenzyme Q
Cyt b
FeS
Cyt c1
Cyt c
Cyt a
Cyt a3
O2
H2O
Complex I
NADH-CoQ
Reductase
FeS
Complex V
ATP synthase
(F0,F1)
FMNH2
NADH+ H+
ADP+Pi
ATP
Substrate

Inhibitors of ETC

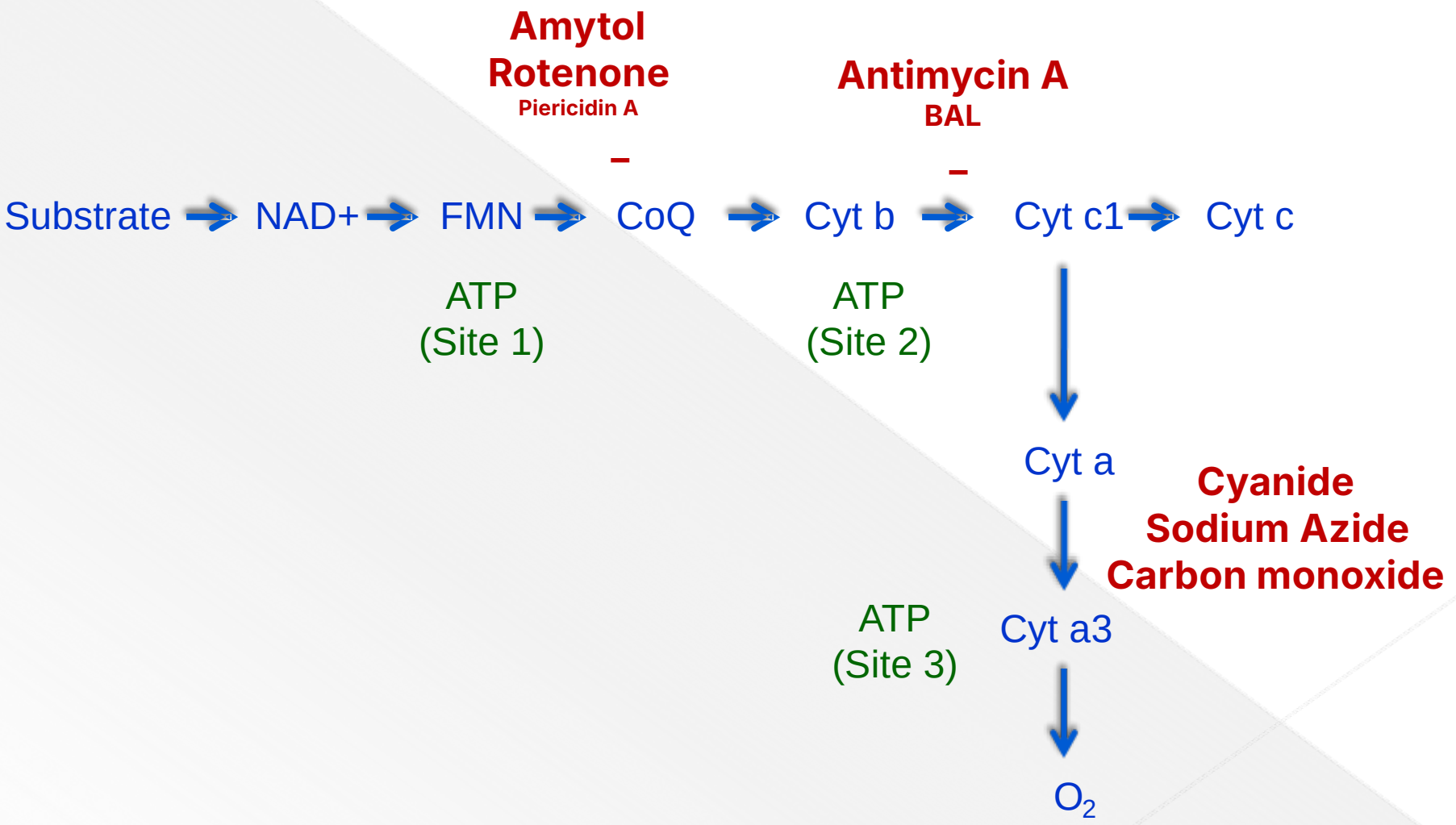
- The inhibitors bind to one of the components of ETC & block the transport of electrons
- This causes the accumulation of reduced components before the inhibitor blockade step & oxidized components after that step.
- The synthesis of ATP is dependent on ETC.

- All the site-specific inhibitors of ETC also inhibit ATP formation.
- NADH & coenzyme Q (Complex I):
- Fish poison rotenone, barbiturate drug amytol & antibiotic piercidin A inhibit this site.
- Complex II: Carboxin inhibit this site.

- **Between cytochrome b & c1 (Complex III):**
- **Antimycin A –an antibiotic, British antilewisite (BAL) –an antidote used against war-gas-Naphthoquinone are important inhibitors of the site between cytochrome b & c1.**
- **Cytochrome oxidase (Complex IV):**
- **Carbon monoxide, cyanide, hydrogen sulphide & azide**

- Effectively inhibit cytochrome while cyanide & azide react with oxidized form of cytochrome.
- Cyanide is most potent inhibitor of ETC
  - It binds to Fe^{3+} of cytochrome oxidase blocking mitochondrial respiration leading to cell death.
- Cyanide poisoning causes death due to tissue asphyxia (mostly of CNS)

Sites of ATP synthesis & Inhibitors



Biological Oxidation & Oxidative Phosphorylation

- **Biological Oxidation:**
- The transfer of electrons from the reduced co-enzymes though the respiratory chain to oxygen is known as **biological oxidation.**
- Energy released during this process is **trapped as ATP.**
- This coupling of oxidation with phosphorylation is called as **OXIDATIVE PHOSPHORYLATION.**

- Complex V of the inner mitochondrial membrane is the site of oxidative phosphorylation.
- Phosphagens act as storage forms of high-energy phosphate and include creatine phosphate, which occurs in vertebrate skeletal muscle, heart, spermatozoa & brain
- Arginine phosphate, in invertebrate muscle.

When ATP is rapidly being utilized as a source of energy for muscular contraction, phosphagens permit its concentrations to be maintained, but when the ATP/ADP ratio is high, their concentration can increase to act as a store of high-energy phosphate.

P:O Ratio

- The P:O ratio refers to the number of inorganic phosphate molecules utilized for ATP generation for every atom of oxygen consumed.
- Approximately P:O ratio represents the number of molecules of ATP synthesized per pair of electrons carried through ETC.

P:O Ratio of 3:

- P/O ratio is 3 for oxidation of substrates producing NADH.
- For each molecule of NADH that is oxidized through ETC 3 ATP are produced.
- Ex: Malate, Pyruvate, Isocitrate, α-Ketoglutarate

P/O ratio of 2:

- P/O ratio is 2 for oxidation of substrates producing $FADH_2$.

- $FADH_2$ transfers electrons to coenzyme Q thus missing the first site of oxidative phosphorylation.
- For each molecule of $FADH_2$ produces 2 ATP.
- Ex: Succinate, fatty acyl CoA, glycerol 3-P.

P/O Ratio of 1:

- P/O ratio is 1 for compounds that transfer electrons to cytochrome oxidase complex.
- Ex: Ascorbic acid.

NOTE:

- Studies on isolated mitochondria indicate P/O ratio of 2.5 for NADH & 1.5 for $FADH_2$

Sites of Oxidative Phosphorylation in ETC

- There are 3 reactions in the ETC that are exergonic,
- Where the energy change is sufficient to drive the synthesis of ATP from ADP and Pi.
  - **Site1:**
  - Oxidation of $FMNH_2$ by coenzyme Q.
  - **Site2:**
- Oxidation of cytochrome b by cytochrome c1
  - **Site3:**
- Cytochrome oxidase.

Energetics of oxidative phosphorylation

- $\frac{1}{2} O_2 + NADH + H^+ \longrightarrow H_2O + NAD^+$
- The redox potential difference between these two redox paires is 1.14V, which is equivalent to an energy 52 Cal/mol
- 3 ATP are synthesized in ETC when NADH is oxidized which equals to 21.9 Cal.

  (each ATP=7.3 Cal)
- The efficiency of energy conservation is calculated as

$$\frac{21.9 \times 100}{52} = 42\%$$

- When NADH is oxidized, about 42% of energy is trapped in the form of 3ATP & remaining is lost as heat.
- The heat liberation is not a wasteful process, since it allows ETC to go on continuously to generate ATP.
- This heat is necessary to maintain body temperature.

Mechanism of oxidative phosporylation

- Two important hypothesis to explain the process of oxidative phosporylation.
- Namely chemical coupling & chemiosmotic
- Chemical coupling hypothesis:
- This hypothesis was put forth by Edward Slater (1953)

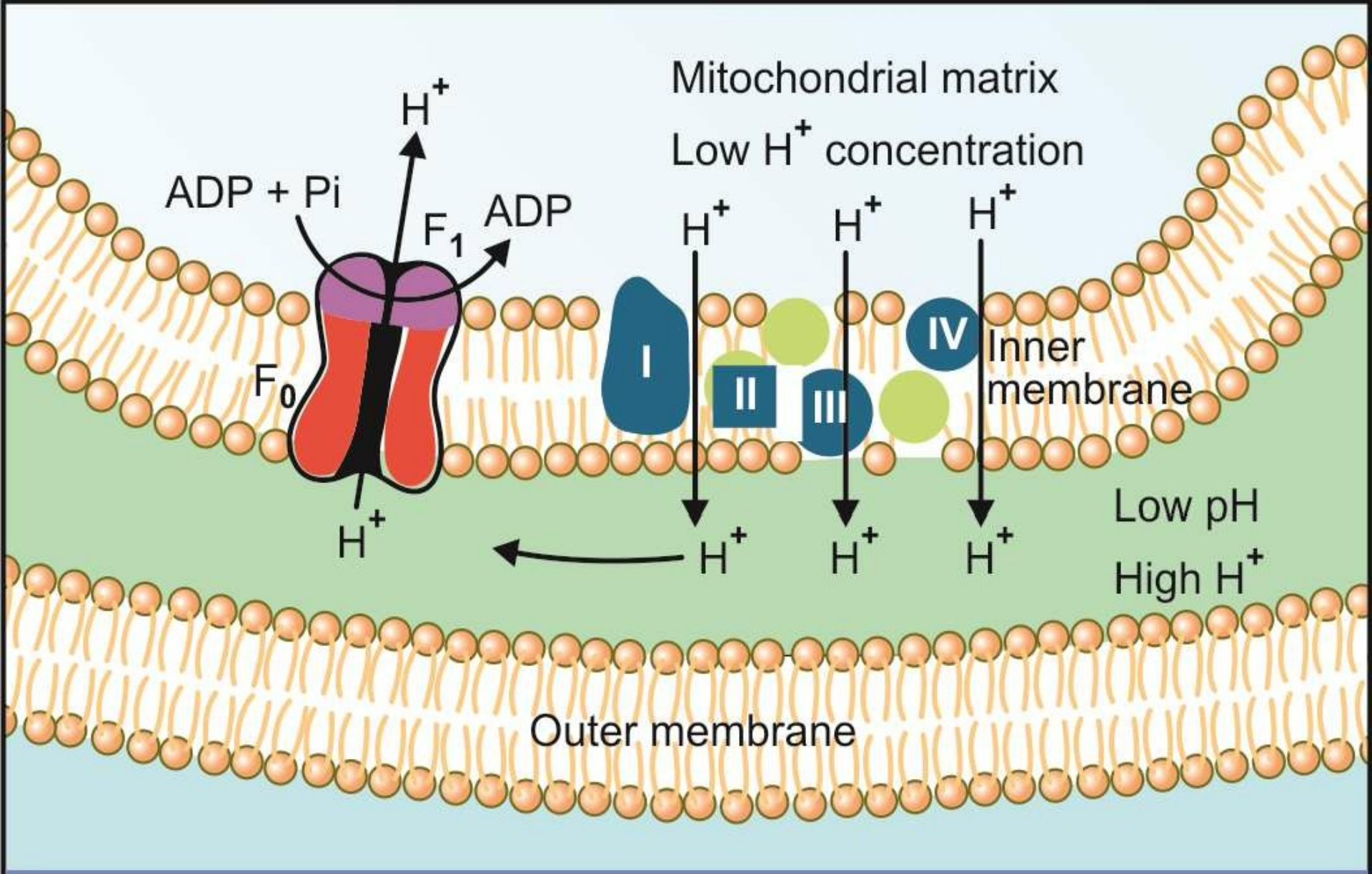
- According to this, during the course of electron transfer in respiratory chain, a series of phosphorylated high-energy intermediates are first produced which are utilized for the synthesis of ATP.
- These reactions are believed to be analogous to the substrate level phosphorylation that occurs in glycolysis or citric acid cycle.
- This hypothesis lacks experimental evidence.

Chemiosmotic Theory

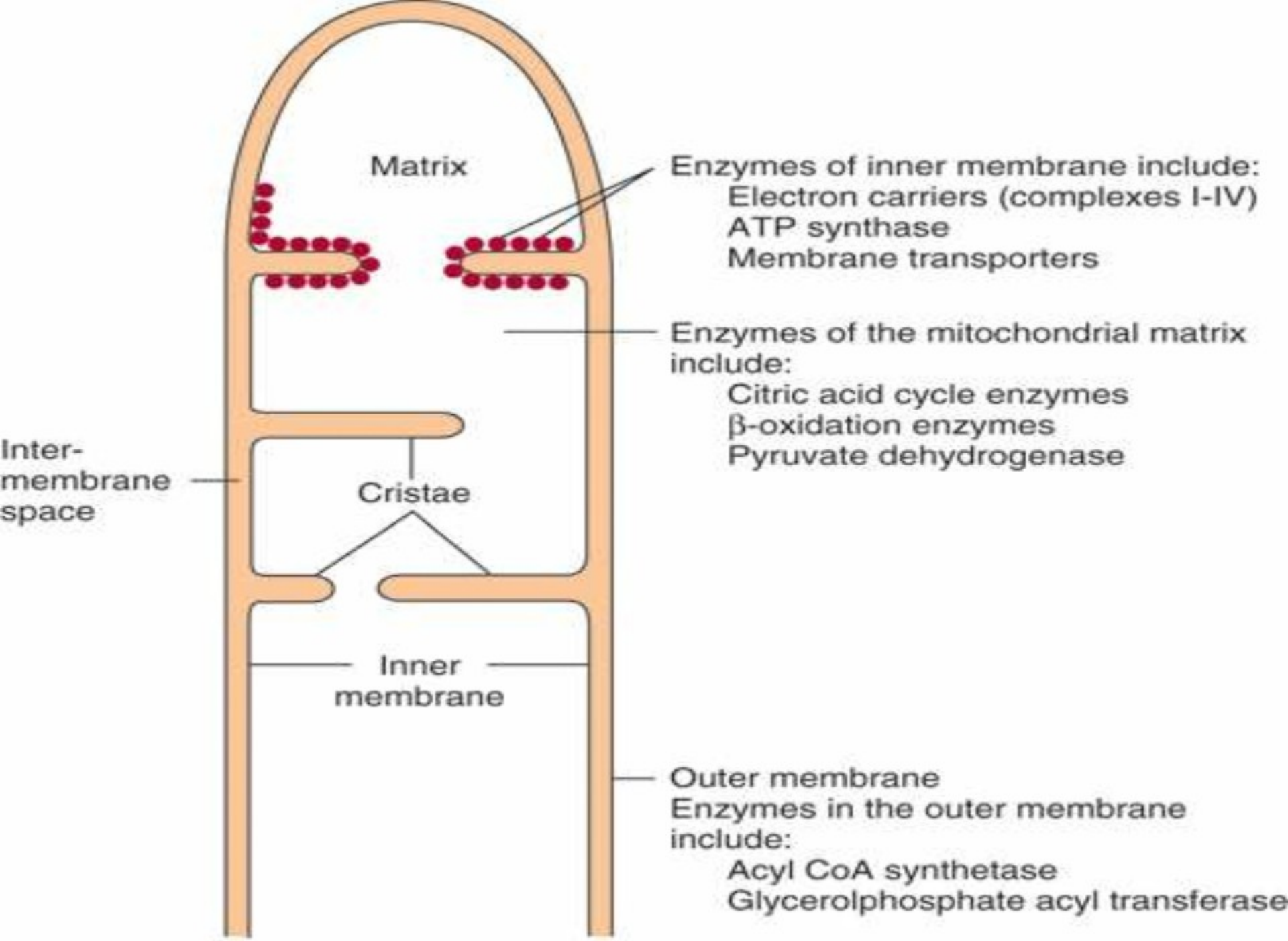
- The transport of electrons through the respiratory chain is effectively utilized to produce ATP from ADP + Pi.
- **Proton gradient:**
- The inner mitochondrial membrane, is impermeable to protons (H^+) & hydroxyl ions (OH^-).

- The transport of electrons through ETC is coupled with the translocation of protons (H^+) across the inner mitochondrial membrane from the matrix to the inter membrane space.
- The pumping of protons results in an electrochemical or proton gradient .

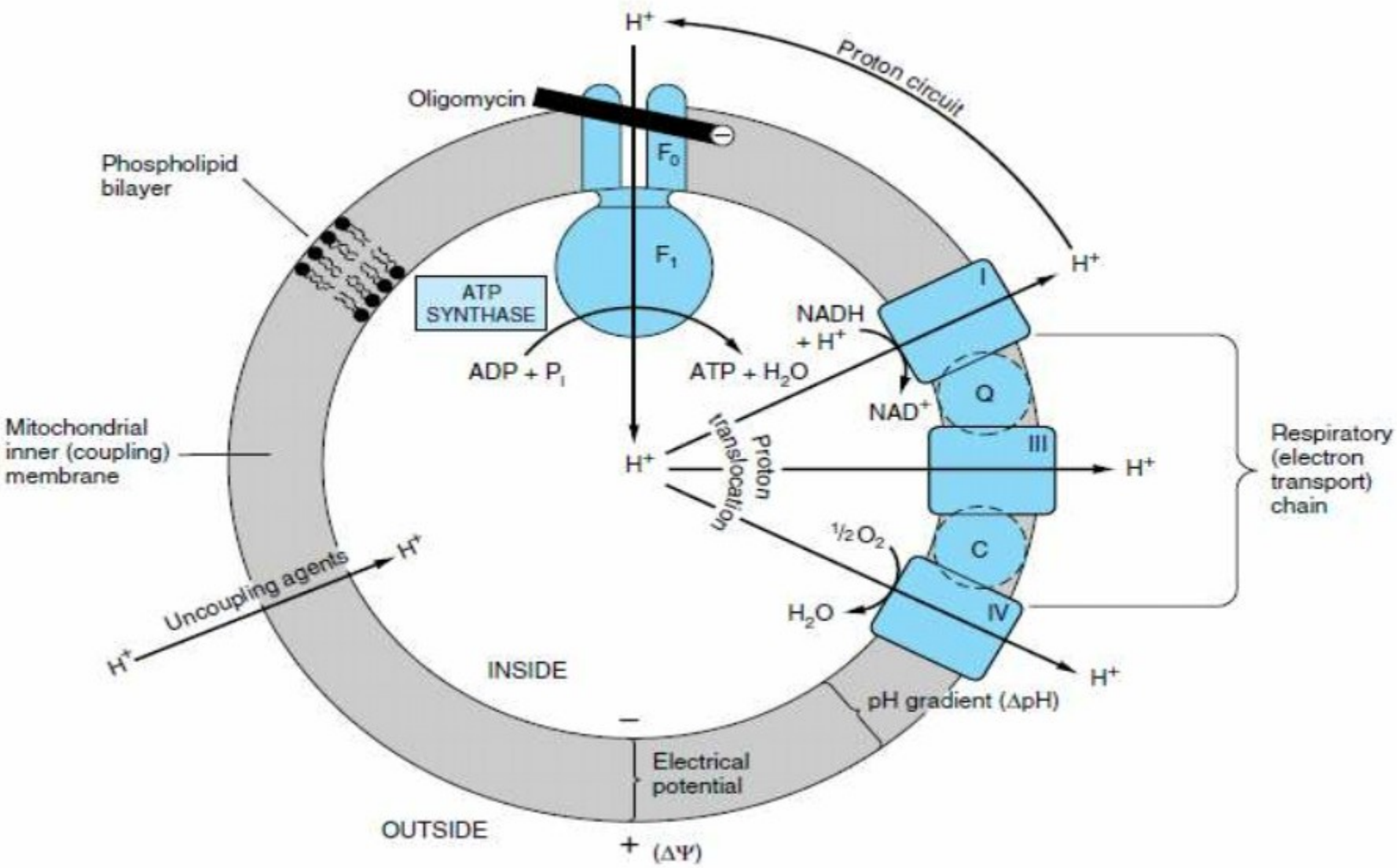
- This is due to the accumulation of more H^+ ions (low pH) on the outer side of the inner mitochondrial membrane than the inner side.
- The proton gradient developed due to the electron flow in the respiratory chain is sufficient to result in the synthesis of ATP from ADP +Pi.



I, II, III, IV = components of ETC; F0, F1, = components of ATP synthase.


Matrix
Enzymes of inner membrane include:
Electron carriers (complexes I-IV)
ATP synthase
Membrane transporters
Enzymes of the mitochondrial matrix include:
Citric acid cycle enzymes
β-oxidation enzymes
Pyruvate dehydrogenase
Inter-membrane space
Cristae
Inner membrane
Outer membrane
Enzymes in the outer membrane include:
Acyl CoA synthetase
Glycerolphosphate acyl transferase

Chemiosmotic Theory



Enzyme systems for ATP synthesis:

- ATP synthase, present in the complex V, utilizes the proton gradient for the synthesis of ATP.
- This enzyme is also known as ATPase, since it can hydrolyze ATP to ADP + Pi.
- ATP synthase is a complex enzyme & consists of two functional subunits, namely F1 & Fo.

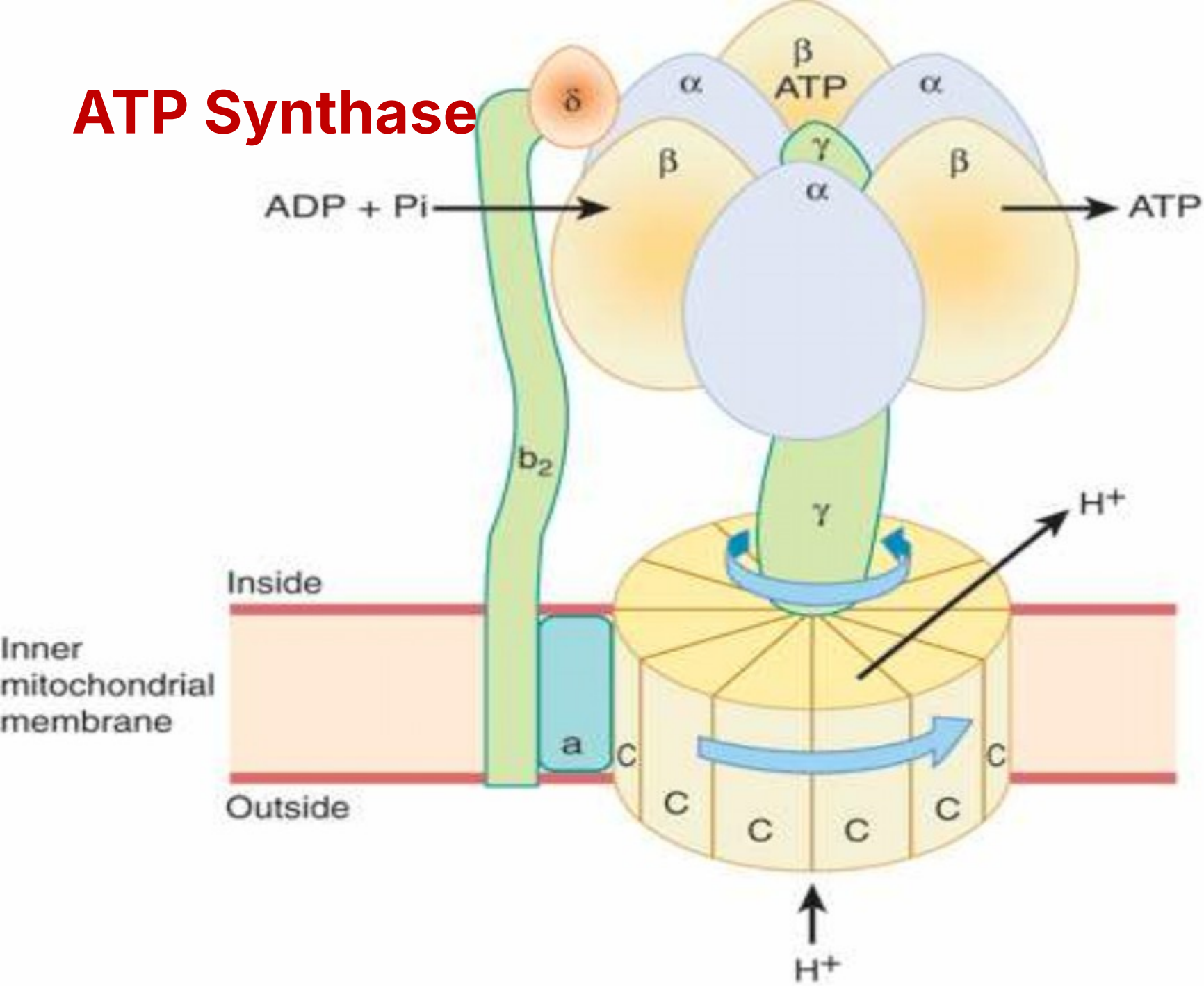
- Fo unit: O stands for oligomycin,
- Fo inhibited by oligomycin.
- Fo spans inner mitochondrial membrane acting as a proton channel through which protons enter the mitochondria
- Fo unit has 4 polypeptide chains & is connected to F1.
- Fo is water insolube whereas F1 is a water soluble peripheral membrane protein.

- F1 unit: It projects into the matrix.
- F1 has 9 polypeptide chains, (3 alpha, 3 beta, 1 gamma, 1 delta, 1 epsilon)
- The α chains have binding sites for ATP & ADP & beta chains have catalytic activity.
- ATP synthesis requires Mg^{+2} Ions.

NO GLYPH
Its structure is comparable with lollipops.

NO GLYPH
The protons that accumulate on the intermembrane space re-enter the mitochondrial matrix leading to the synthesis of ATP.

ATP Synthase



The Nobel Prize in Chemistry 1997

Press release

"for their elucidation of the enzymatic mechanism underlying the synthesis of adenosine triphosphate (ATP)"

Paul D. Boyer

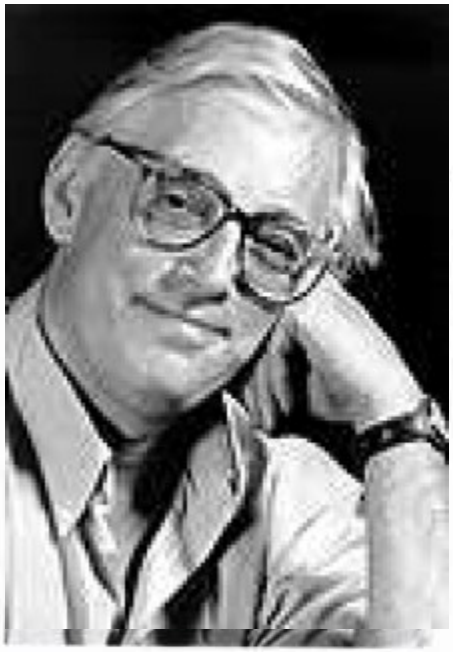
USA

University of California, Los Angeles, USA

1918 -

Autobiography
Nobel Diploma
Prize Award
Ceremony





John E. Walker

U.K.

Medical Research Council Laboratory of Molecular Biology, Cambridge, U.K.

1941 -

Autobiography
Nobel Diploma
Prize Award
Ceremony

Rotor motor model for ATP generation

- Paul Boyer in 1964 proposed that a conformational change in the mitochondrial membrane proteins leads to the synthesis of ATP
- This is now considered as rotary motor/engine driving model or binding change model, is widely accepted for the generation of ATP.

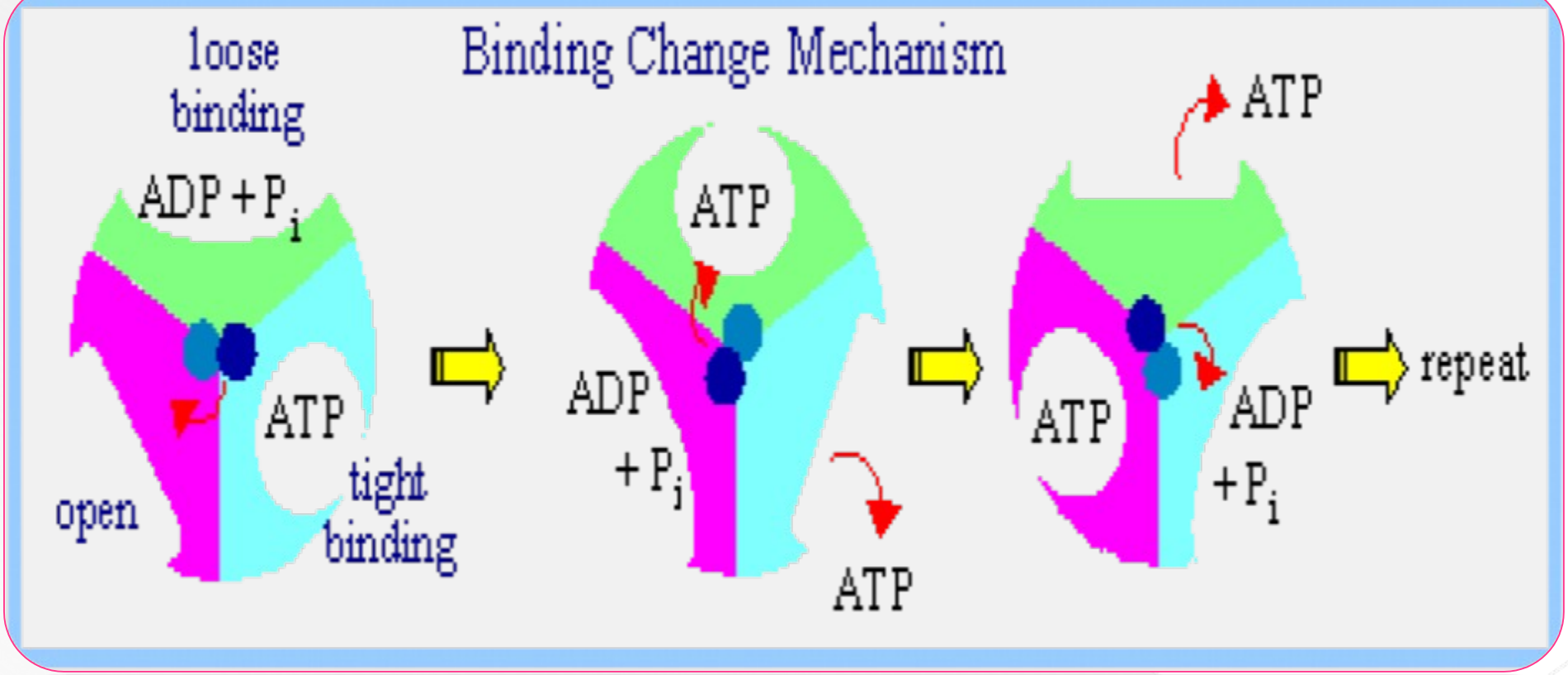
- The enzyme ATP synthase is Fo & F1 complex
- The Fo sub complex is composed of channel protein ‘C’ subunits to which F1-ATP synthase is attached.
- F1-ATP synthase consists of a central gamma-subunit surrounded by alternating alpha & beta subunits (α3 & β3).
- In response to the proton flux, the gamma subunit physically rotates.

- This induces conformational changes in the β3 subunits that finally lead to the release of ATP.
- According to the binding change mechanism, the three β subunits of F1 - ATP synthase adopt different conformations.
- One subunit has Open (O) conformation, the second has loose (L) conformation while the third one has tight (T) conformation.

- By an known mechanism, protons induce the rotation of gamma subunit, which in turn induces conformation changes in β subunits,.
- The substrates ADP & Pi bind to β subunit in L conformation.
- The L site changes to T conformation, & this leads to the synthesis of ATP.

- The O site changes to L conformation which binds to ADP + Pi.
- The T site changes to O conformation & releases ATP.
- This cycle of conformation changes of β subunits is repeated.
- Three ATP are generated for each revolution.

Binding chain mechanism



- Protons entering the system, cause conformational changes in F1 particle.
- The 3 beta subunits are in three functional states, O (open), L (loose) & T(tight).
- Conformational change induces catalytic activity.
- Open form is regained after release of ATP.

Inhibitors of Oxidative phosphorylation

- The mitochondrial transport of electrons is tightly coupled with oxidative phosphorylation.
- Oxidation & phosphorylation proceed simultaneously.
- There are certain compounds that can uncouple (or delink) the electron transport from oxidative phosphorylation.

- Such compounds are known as uncouplers, increase in the permeability of inner mitochondrial membrane to protons (H^+).
- The result is that ATP synthesis does not occur
- The energy linked with the transport of electrons is dissipated as HEAT.
- The uncouplers allow (often at accelerated rate) oxidation of substrates (via NADH or $FADH_2$) without ATP formation.

Examples:

2,4-dinitrophenol (DNP):

- It is small lipophilic molecule.
- DNP is a proton – carrier & easily diffuse through the inner mitochondrial membrane.
- Others –dinitrocressol, pentachlorophenol, trifluorocarbonylcyanide, phenylhydrazone

Physiological uncouplers

- Certain physiological substances which act as uncouplers at higher concentration.
- These are thermogenin, thyroxine and long chain fatty acids & unconjugated bilirubin

Significance of uncoupling:

- The maintenance of body temperature is particularly important in hairless animals, hibernating animals & the animals adopted to cold
- These animals possess a specialized tissue called brown adipose tissue in the upper back & neck portions.

- The mitochondria of brown adipose tissue are rich in electron carriers & are specialized to carry out an oxidation uncoupled from phosphorylation.
- This causes liberation of heat when fat is oxidized in the brown adipose tissue.
- The presence of brown adipose tissue in certain individuals is believed to protect them from becoming obese.

- **Thermogenin** is a natural uncoupler located in the inner mitochondrial membrane of brown adipose tissue
- It acts like an uncoupler, blocks the formation of ATP, & liberates heat.
- **Ionophores:** These are lipophilic substances that promote the transport of ions across biological membranes.
- **Valinomycin & nigercin** also act as uncouplers.

Inhibitors of oxidative phosphorylation

- **Oligomycin:** This antibiotic prevents the mitochondrial oxidation as well as phosphorylation.
- It binds with enzyme ATP synthase & blocks the proton(H^+) channels.
- Thus it prevents the translocation (re-entry) of protons into the mitochondrial matrix.

- Due to this, protons get accumulated at higher concentration in the inter membrane space
- Electron transport is stoped.
- **Atractyloside:** It is a plant toxin & inhibits oxidative phosphorylation.
- It blocks the adequate supply of ADP.

Inherited disorders of oxidative phosphorylation

- 100 polypeptides are required for oxidative phosphorylation.
- Of these, 13 are coded by mitochondrial DNA & synthesized in the mitochondria, while the rest are produced in the cytosol (coded by nuclear DNA) & transported.
- mtDNA is maternally inherited since mitochondria from the sperm do not enter the fertilized ovum.

- Mitochondrial DNA is 10 times more susceptible to mutations than nuclear DNA.
- mtDNA mutations are commonly seen in tissues with high rate of oxidative phosphorylation (e.g. CNS, skeletal & heart muscle, liver).

Diseases:

- Lethal infantile mitochondrial opthalmoplegia
- Leber’s hereditary optic neuropathy (LHON)
- Myoclonic epilepsy
- Mitochondrial encephalopathy lactic acidosis stroke like episodes (MELAS)

Oxidative Phosphorylation Diseases

Syndrome	Feature
Laber's heriditory Optic neuropathy (LHON)	Complex I defect, Blindness, cardiac conduction defects.
Myoclonic epilepsy ragged red fiber disease (MERRF)	Myoclonic epilepsy, myopathy, dementia.
Mitochondrial encephalopathy lactic acidosis stroke like episodes (MELAS)	Complex I defect; Lactic acidosis, stroke, myopathy, dementia.
Leigh's syndrome	Complex I defect, Movement disorders.

References

- Lippincott
- Harper
- Chatterjea
- Lehninger